# Digital Media Formats & Compression

## Understanding Image, Audio, and Video Quality for Web and Screen

**Savannah Longmore**

DWDD 2420

Michael Harper

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# TABLE OF CONTENTS


**Table of Contents.....2**

**Introduction.....3**

**Chapter 1: Technology Assessment.....4**

Data Storage.....5
Lossy vs Lossless.....7
Vector vs Raster.....8
Aliasing vs Anti-Aliasing.....9

**Chapter 2: Image Compression.....10**

Image Formats.....11
JPEG: Save A Copy.....12
JPEG: Save As.....13
JPEG: Save for Web.....14
JPEG: Comparison.....15
Squoosh: JPEG.....16
Squoosh: WebP.....17
Squoosh: AVIF.....18
Squoosh: Comparison.....19
Photoshop: WebP.....20
Photoshop: Format Comparison.....21

**Chapter 3: Audio Format.....23**

Audio Format.....24
Audo: Intro.....25
Audio: Streaming.....26

**Chapter 4: Video Format.....28**

Video Format.....29
Video: Intro.....30
Video: Streaming.....31
Video: W/ Levono.....32
Video: W/ Pioneer.....33
Video: Closer Look.....34

**Asset Management Summary.....36**

**Conclusion.....37**

**Glossary.....38**
**Works Cited.....39**

# Introduction

Digital media is everywhere—on our phones, laptops, televisions, and across every website and streaming platform we use daily. As designers, it is essential to understand how images, audio, and video are created, stored, compressed, and displayed. The technical choices made behind the scenes directly affect visual quality, audio clarity, file size, loading speed, and overall user experience.

This project explores the core technical concepts that shape digital media, including screen resolution, aspect ratio, bit depth, vector and raster graphics, image compression, audio formats, and video formats. Through hands-on testing in programs such as Photoshop and Squoosh, as well as real-world comparisons between streaming platforms like Disney+, Amazon Prime Video, and Netflix, this project examines how quality changes as media is compressed, exported, and displayed on different devices.

By visually analyzing compression artifacts, pixel structure, color depth, and audio fidelity, this project demonstrates how small technical decisions can create large differences in perceived quality. Ultimately, the goal of this research is to help designers make informed choices that balance file size, performance, and visual or audio integrity—ensuring professional, efficient, and high-quality digital experiences across all platforms.

# 01 Technology Assessment

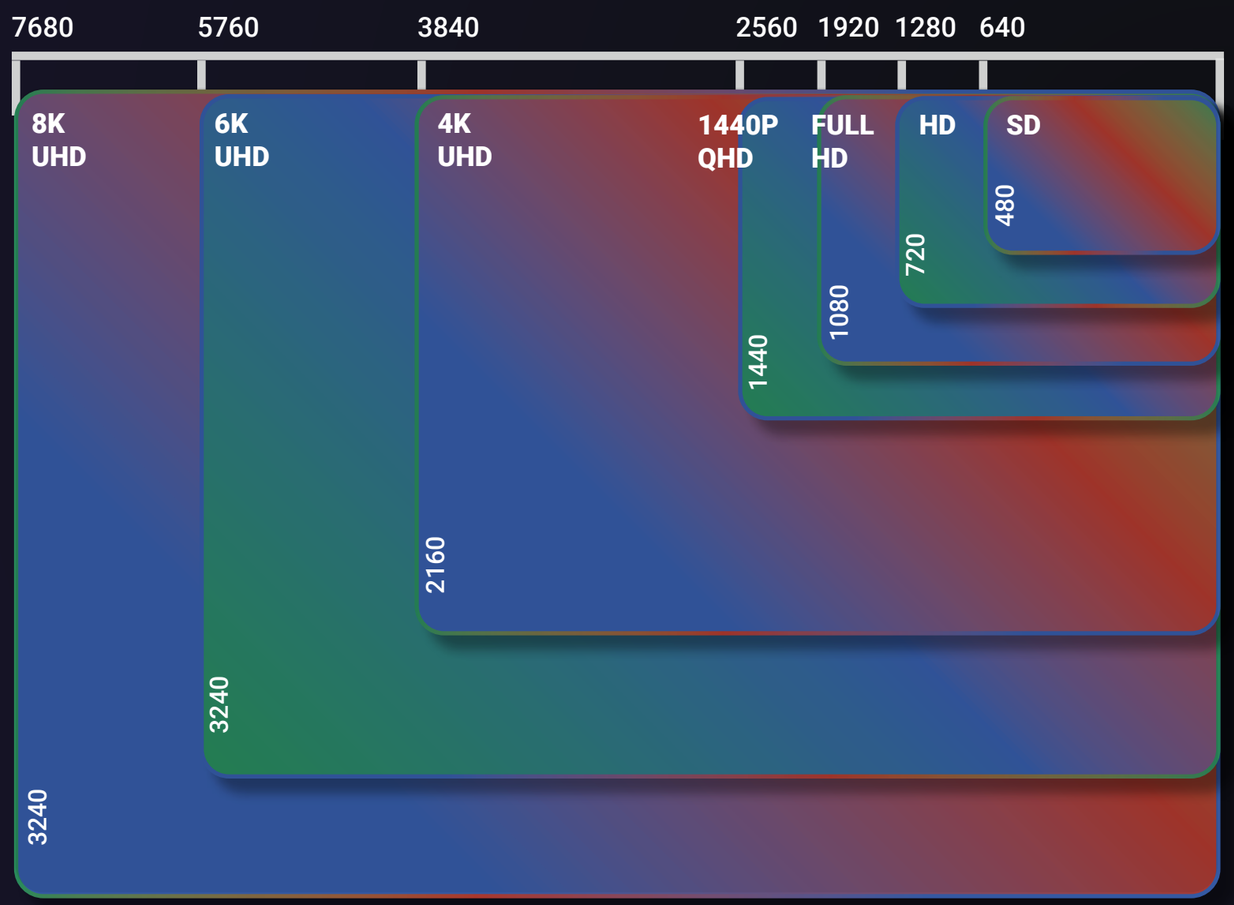
## Overview

Bit depth and resolution play a critical role in digital design, especially in web-based environments where image clarity, performance, and file size must be carefully balanced. Many factors contribute to visual artifacts and reduced image quality, making it essential to understand how these issues occur and how to prevent them. In this chapter, we will explore bit depth and screen resolution, as well as lossy versus lossless compression, vector versus raster graphics, and aliasing versus anti-aliasing.

# DATA STORAGE

## Screen Resolution

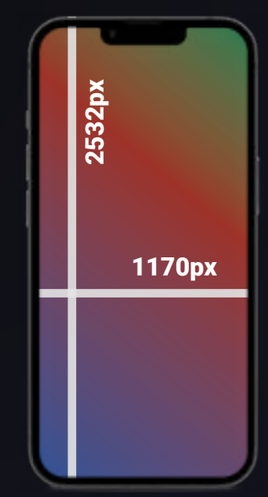
Screen resolution refers to the number of pixels that can be displayed on a screen, whether it's horizontally or vertically. Pixels can either be shown as squares or circles, and the more PPI (Pixels Per Inch), the clearer the image will come across or be displayed. If the image becomes stretched or scaled up, the pixels are more noticeable, but if the image becomes smaller or shrinks, the pixels become sharper to create a better quality picture. While resolution defines how many pixels are displayed, aspect ratio determines how those pixels are proportionally arranged on the screen.




## Aspect Ratio

When it comes to the aspect ratio, for every height and width, several pixels make up the resolution. For example, if the ratio is 16:9, for every 16 pixels in width, there are 9 pixels in height. So if the screen resolution is 1920x1080, the aspect ratio would be 16:9. The higher the number, the more detailed the picture can be.

## Device Resolution Examples




**iPhone 14**

Resolution 2532 x 1170 px

Aspect Ratio: 19.5:9

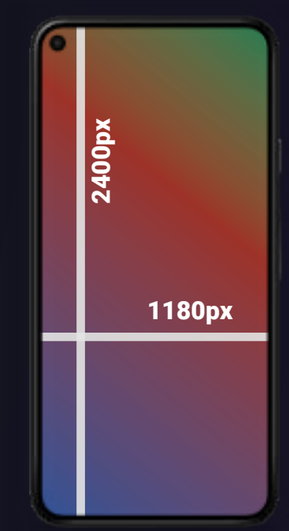
PPI: Approximately 460

**Google Pixel 5**

Resolution 2340 x 1080px

Aspect Ratio: 19.5:9

PPI: Approximately 432




**Samsung Galaxy S21+**

Resolution 2400 x 1080px

Aspect Ratio: 20:9

PPI: Approximately 394

# DATA STORAGE

## Data Storage Size

Data storage is defined using metric prefixes, starting at bits. When the number of bits increases to 1,024, the word Bit becomes Bytes, and when Bytes increase again to 1,024, the world changes to Kilobytes, and so on. If we think about the letter “z”, this letter takes up a whole Bit of data storage. Higher bit depth increases the amount of color data stored per pixel, which often contributes to larger file sizes when combined with high resolution. (Study).

**Data Storage Size**

| Unit | Capacity |
|---|---|
| Bit (b) | 1 or 0 (on/off) |
| Byte (B) | 8 bits |
| Kilobyte(KB) | 1024 Bytes |
| Megabyte (MB) | 1024 Kilobytes |
| Gigabyte (GB) | 1024 Megabytes |
| Terabyte (TB) | 1024 Gigabytes |
| Petabyte (PB) | 1024 Terabytes |
| Exabyte (EB) | 1024 Petabytes |
| Zettabyte (ZB) | 1024 Exabytes |
| Yottabyte (YB) | 1024 Zettabytes |

**Bit Depth**

| Bits Per Pixel | Colors Available |
|---|---|
| 1 | 2 |
| 2 | 4 |
| 4 | 16 |
| 8 | 256 |
| 16 | 65,536 |
| 24 | 16,777,216 |
| 32 | 16,777,216 + Transparency |

## Bit Depth

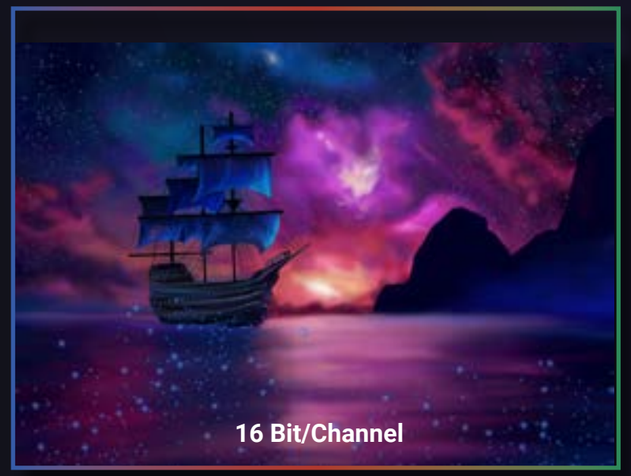
Bit depth (or color depth) is defined as the number of bits used to represent the color of each pixel, which determines the total number of unique colors an image can potentially contain. If an image has a higher bit depth, the transitions between colors are significantly smoother, making gradients softer and more blended. This precision is built on the fact that each color channel uses a binary scale; for example, in an 8-bit image, a channel has 28 = 256 possible values (from 0, which is no intensity, to 255, which is maximum intensity) (Imagekit).

Example:

256 x 256 x 256 = 16,777,216

The picture to the right displays the bit depth of 8. Because the bit depth is so small, we start to see sections that begin to clutter together as information is being extracted and removed from the image.

From the next image, we can see that it’s much clearer and crisper. This shows minimal visible information loss, indicating that 16-bit depth preserves significantly more detail than 8-bit.




Lastly, we have a 32-bit depth image. Comparing 16 to 32, we can’t see much difference. There are no sections where the pixels are being combined and causing abstraction. Overall, it’s just as crisp and clear as the 16-bit depth image.

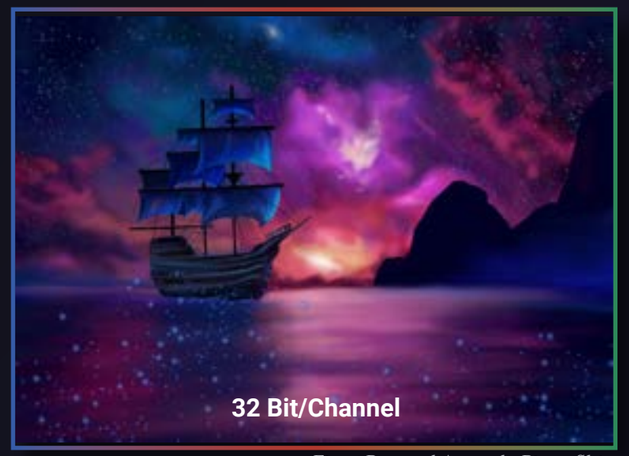



Fig. 5 Personal Artwork, Pirate Ship

# LOSSY vs. LOSSLESS

## Lossy

Lossy compression reduces file size by discarding some image data, prioritizing smaller storage over perfect fidelity.

This makes lossy images ideal for websites and other digital media where file size matters, but enlarging them can cause pixelation, artifacting, and graininess (Adobe).

## Lossless

In contrast, lossless compressions preserve all original image data, resulting in higher-quality images that remain sharp even when scaled up or down. These files are larger but are ideal for logos, icons, or images that need to retain full detail (Adobe).

### Lossy Files

JPEG

HEIC

WebP

Fig. 6 Adobe, Lossy Image,

### Lossless Files

PNG

BMP

RAW

Fig. 7 Adobe, Lossless Image,

# VECTOR vs. RASTER

## Vector

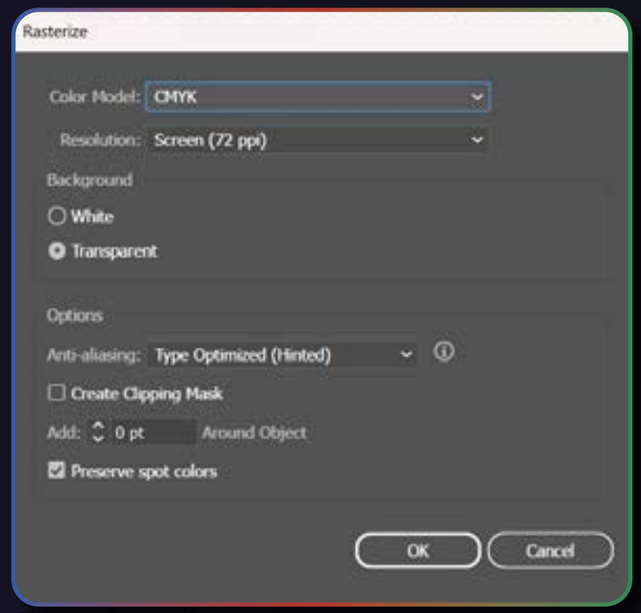
Vector images are created using mathematical formulas to define shapes, lines, curves, and colors. This allows them to scale up or down indefinitely without any loss of quality, making them ideal for logos, icons, typography, and illustrations. Vector images are resolution-independent and maintain crisp edges at any size. While vectors excel at scalability and precision, they are less suited for highly detailed, photo-realistic images compared to raster graphics.

**Real-World Examples:** Commonly used for logos, icons, maps, and typography.

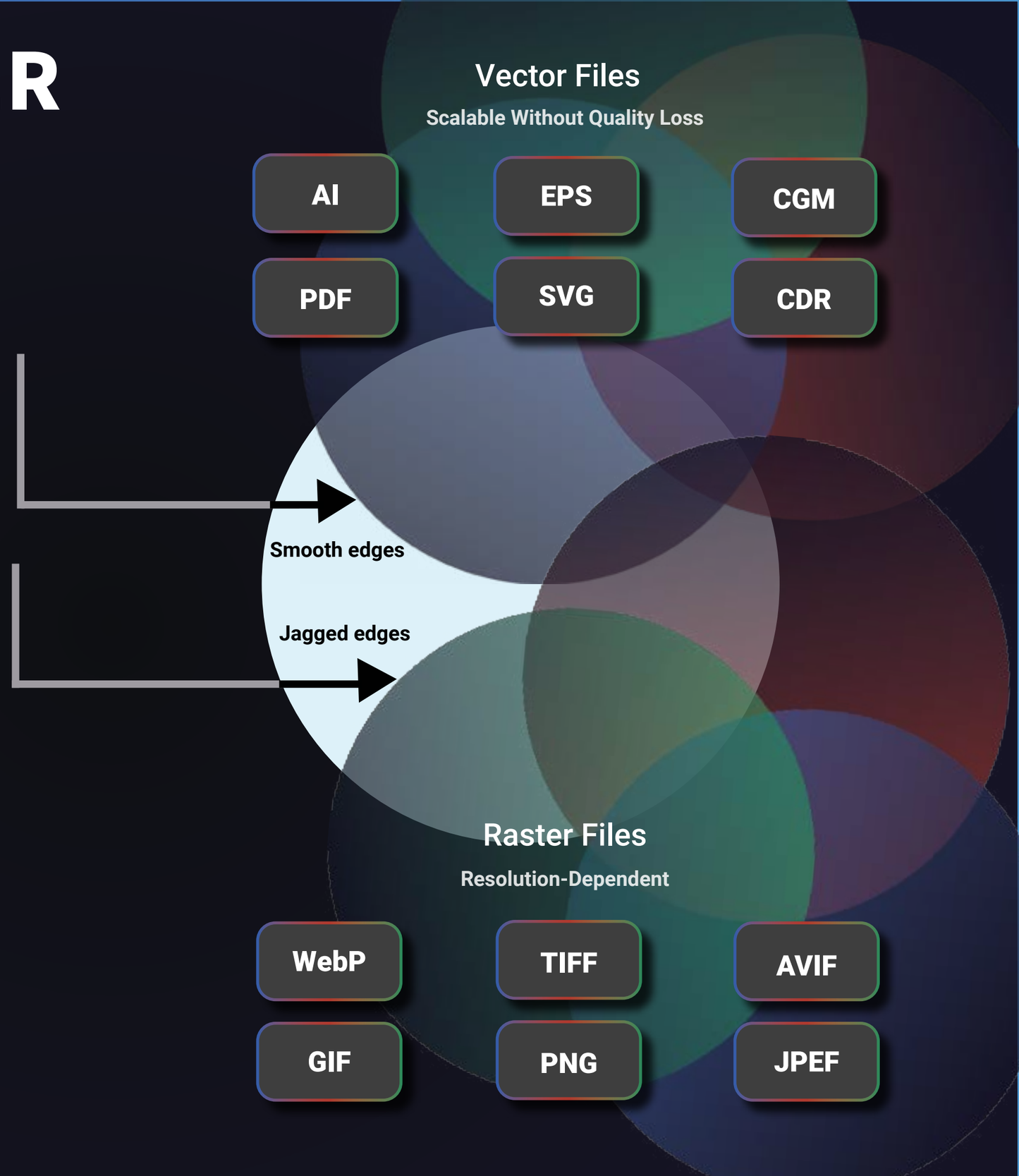
## Raster

Raster images are two-dimensional graphics made up of a grid of individual pixels, each storing a specific color or brightness value. They are ideal for photographs, digital paintings, and other images that require rich detail and smooth color transitions. Because raster images are pixel-based, scaling them up beyond their original size can cause pixelation and blurriness.

**Real-World Examples:** Commonly used for photos, digital artwork, and web images.




Illustrator Rasterize Settings

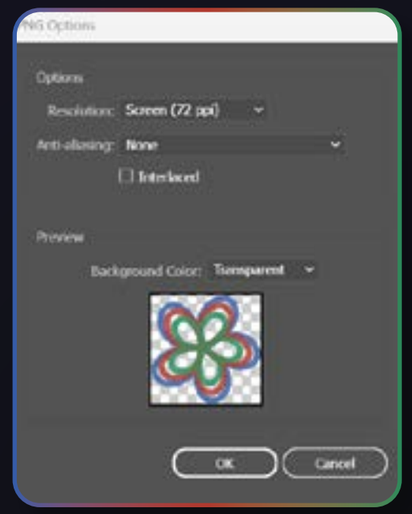


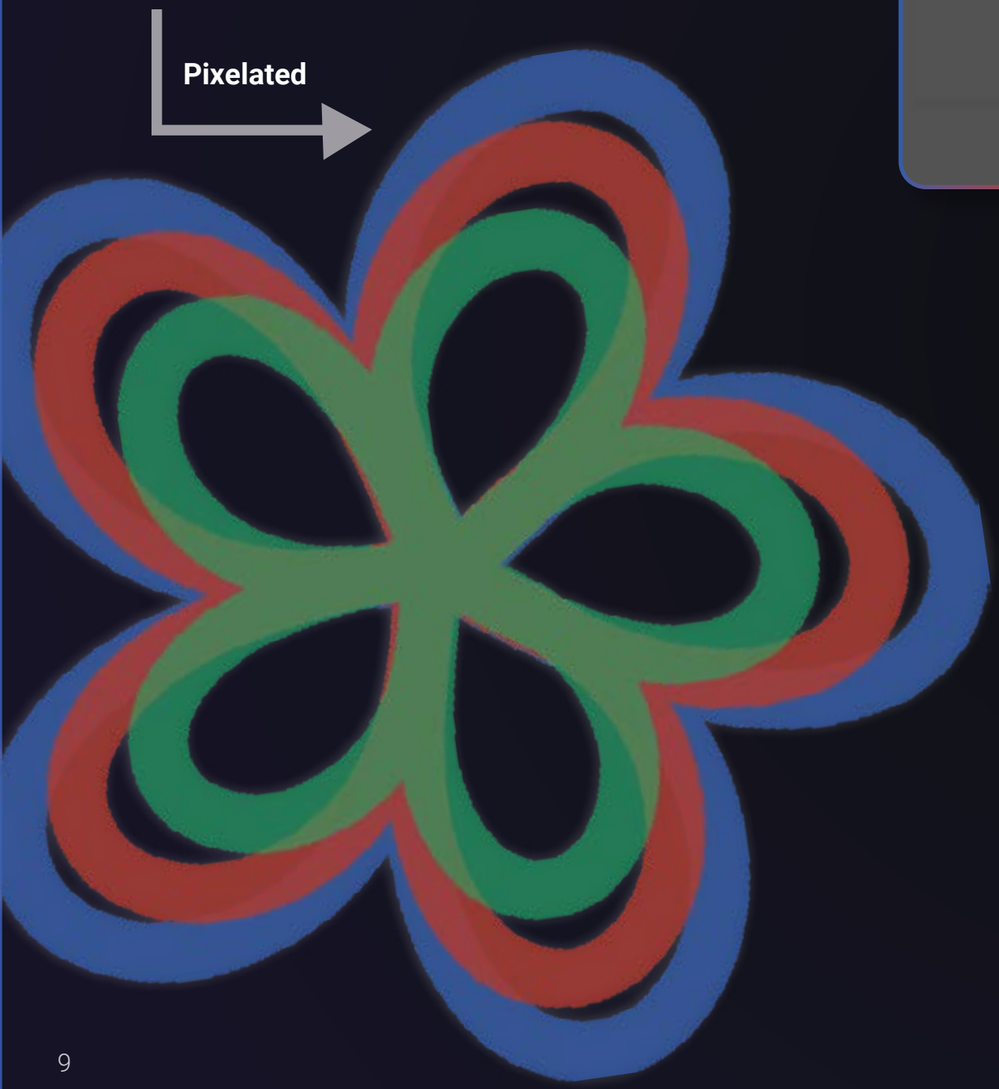
# ALIASING vs. ANTI-ALIASING

## Aliasing

Aliasing occurs when an image or line is displayed at a resolution too low to accurately represent its detail. This produces visible jagged, stair-step edges known as "jaggies." Aliasing can also introduce pixelation, color noise, and Moiré patterns.

Alias Export Settings (No Anti-Aliasing)

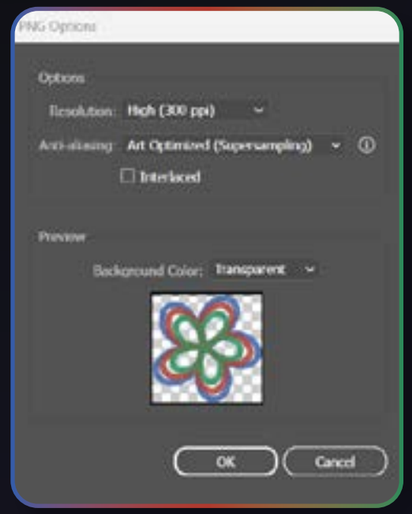


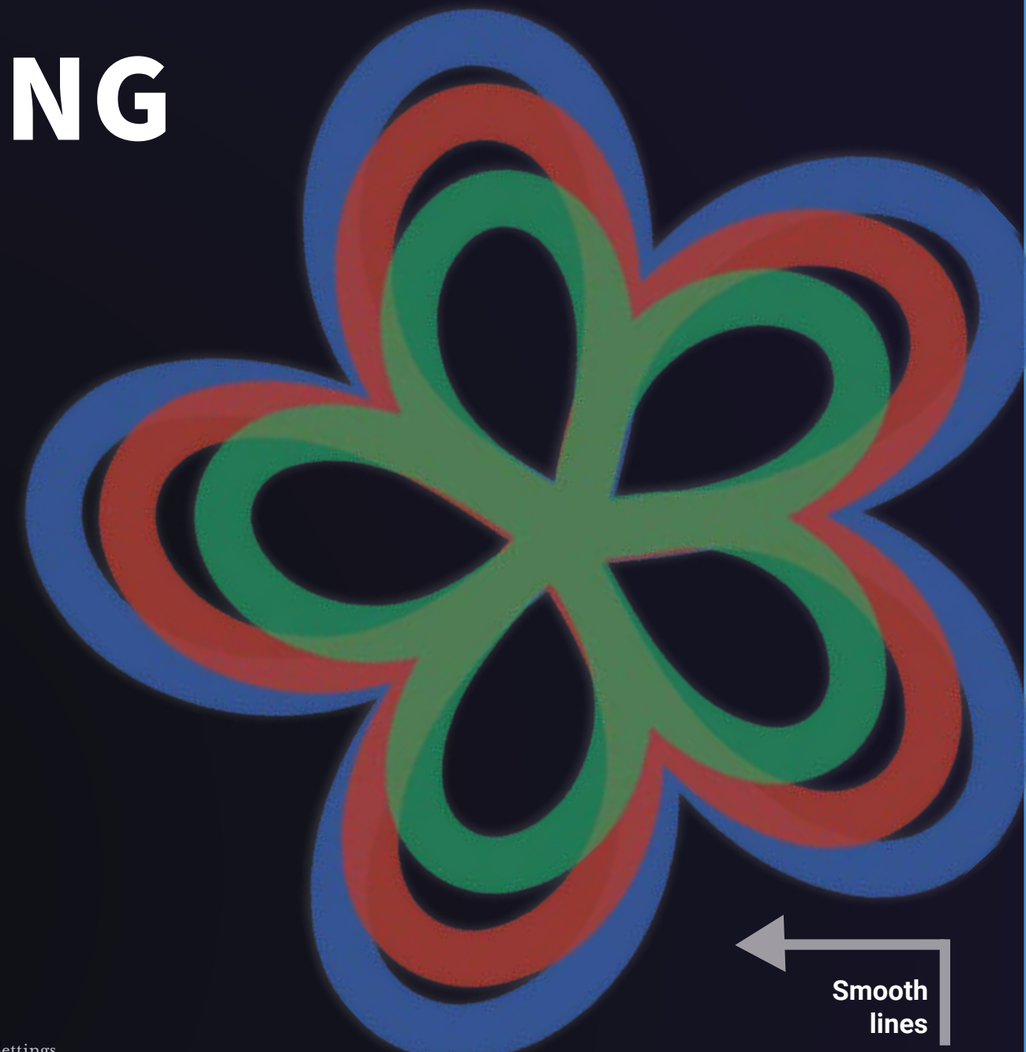




JPEG

PNG

Anti-Aliased Export Settings




## Anti-aliasing

Anti-aliasing is a smoothing technique that reduces the appearance of jagged edges by blending edge pixels with surrounding colors. This creates the illusion of smoother, cleaner lines and higher visual quality.

**Key Difference:** Anti-aliasing blends edge pixels to reduce stair-stepping, while aliasing preserves hard pixel boundaries, resulting in jagged edges.

# 02 Image Compression

## Overview

Digital images form the foundation of modern visual media across websites, applications, and digital design. Image formats determine how visual data is stored, compressed, and displayed on different devices and screens. The way an image is encoded directly affects file size, clarity, color depth, and loading performance. In this section, we will examine how adjusting JPEG compression quality in Photoshop affects perceived image quality and file size. We will also explore how Squoosh compresses images using formats such as JPEG, WebP, and AVIF, and compare how each compression method impacts image quality and resolution.

# IMAGE FORMATS

## Intro

Though many different image files can be used, there will only be a select few that we will really dive into. For now, we will go over the quick facts of JPEG, PNG, WebP, and HEIC.

## .JPEG (.jpg .jpeg .jpe .jif .jfif .jfi)

**Stands for:** Joint Photographic Experts Group
**Created:** 1995
**Compression:** Lossy
**History:** International Organization for Standardization (ISO)wanted a way to bring photo-realistic pictures to small screens around the world.
**Facts:**

- Supports up to 24-bit color
- Compatible with most browsers, software, and apps
- Allows for quick and fast access to viewing online
- Uses compression models that remove image data that are less likely to be perceived
- JPEGs are smaller formats than GIFs
- The quality will suffer with compressed images, and lines and edges will lose their sharpness
- Losing a lot of information will cause aliasing on edges, blooming, or noise.

## .PNG

**Stands for:** Portable Network Graphic
**Created:** Late 1980s and early 1990s
**Compression:** Lossless
**History:** The IT-expert Oliver Fromme came up with the name PING, later shortened to PNG. PNGs were made to support a large veriety of color.
**Facts:**

- Retains all of its original data when compressed
- Good for detailed graphics and charts on a website
- PNG can store many more details than GIFs
- PNGs can handle millions of colors instead of thousands
- PNG is an open format and can be edited in any program
- PNG is a type of raster image file
- Easier to store and transfer
- PNG takes up more storage space
- PNGs don't support CMYK color modes
- Transferring PNG to print can be difficult
- Larger file size means slower loading times

## .WebP

**Stands for:** Web Pictures
**Created:** 2010
**Compression:** Lossy and Lossless
**History:** Google launched the WebP format as part of its mission to make loading times faster across the internet.
**Facts:**

- Format is lighter than JPEG and PNG
- Higher image quality with faster loading
- Can support transparency and animation
- Less usual for viewing images offline
- Old browsers like Edge and Firefox don't support WebP
- Sometimes, WebP compression can degrade the image quality
- Limited support in some graphic editing software

## .HEIC (.HEIF)

**Stands for:** High Efficiency Image Container
**Created:** Created in 2015 and was adoted in 2017
**Compression:** Lossy and Lossless
**History:** used by Apple across its mobile devices and is an updated variant of the High Efficiency Image Format (HEIF).
**Facts:**

- HEIC files are half the size of JPEGs
- Can hold multiple images in one file
- Takes up less storage
- HEIC files still offer the same high-quality images despite being smaller than other image file types
- Supports 16-bit color
- HEIC retains important data
- HEICs store edits and can revert to the original
- HEIC hasn't gained widespread usage outside the Apple ecosystem
- Some platforms aren't HEIC compatible

## .AVIF (.AVIF)

**Stands for:** AV1 Image File Format
**Created:** 2019
**Compression:** Primarily lossy but supports Lossless
**History:** AVIF is maintained by Alliance for Open Media (AOMedia) which maintains the AV1 codec and ISO/IEC through the ISO Base Media File Format (ISO/IEC 14496-12).
**Facts:**

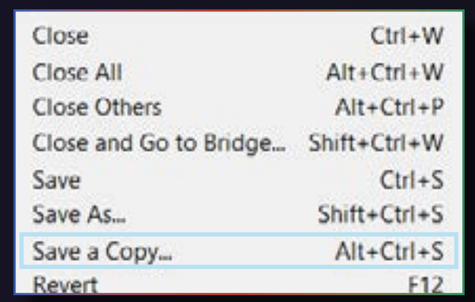
- AVIF image format based on the AV1 video codec and the ISO Base Media File Format (ISOBMFF)
- AVIF stores: AV1-compressed image data, Metadata (EXIF, XMP), HDR and color profiles
- AV1 video codec is developed by the Alliance for Open Media as a royalty-free replacement for VP9 and HEVC.
- AVIF often achieves significantly smaller file sizes than JPEG at comparable visual qualityAVIF supports 8-bit, 10-bit, and 12-bit color
- AVIF is now considered a next-generation image standard, competing with: JPEG, PNG, WebP, and HEIC

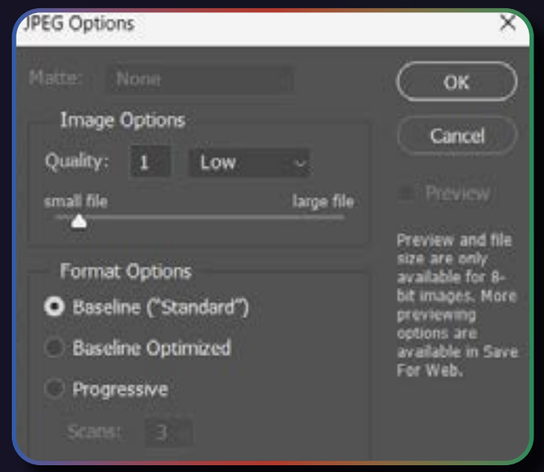
# JPEG: SAVE A COPY

## Save a Copy (SAC):

When comparing JPEG quality 10 to quality 5, there is very little visible difference across the overall image. Even in the close-up views, both versions retain most of their detail, although the quality 5 image begins to show early signs of blocking and compression artifacts. Visually, the two images appear nearly identical, with the most significant difference being the file size. This is an important consideration for designers, particularly when creating content for websites or uploading images to digital platforms, where balancing quality and file size is essential.

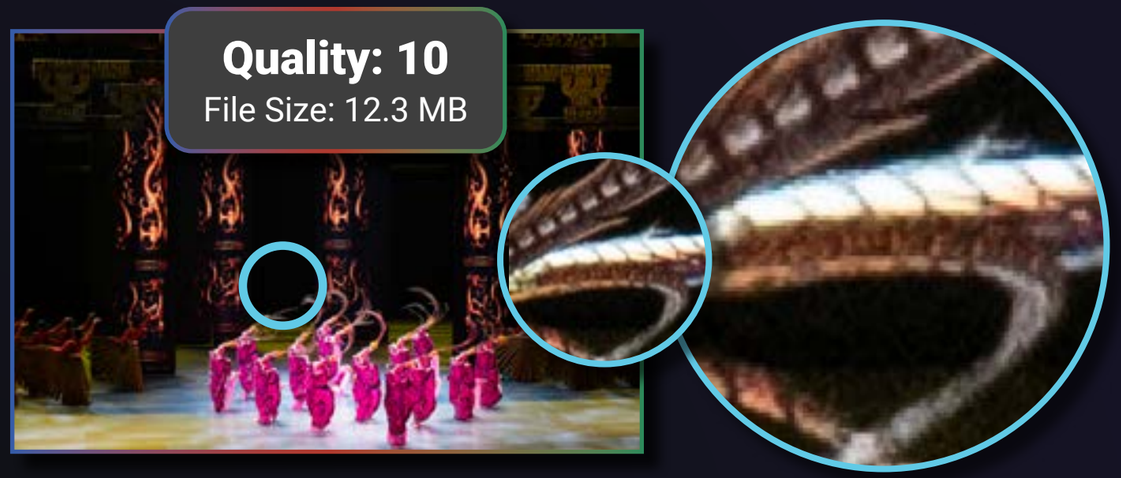
In contrast, when examining the image at quality 1, a clear degradation in visual quality becomes apparent. At this level, heavy blocking and pixelation dominate the image. Although these artifacts may be less noticeable when viewed from a distance, the loss of detail is significant. This comparison highlights how lower compression settings can dramatically alter an image's appearance and emphasizes the importance of selecting appropriate export settings based on the intended use of the image.

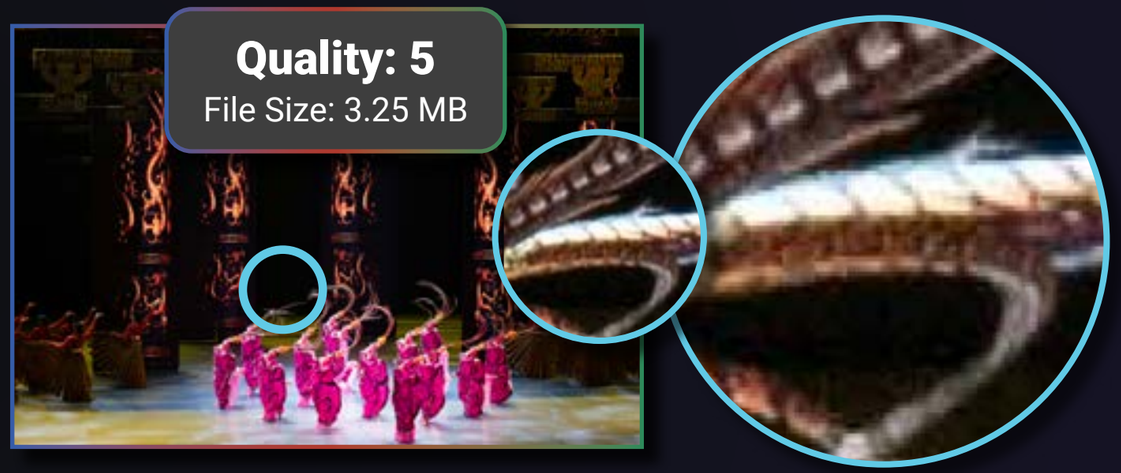


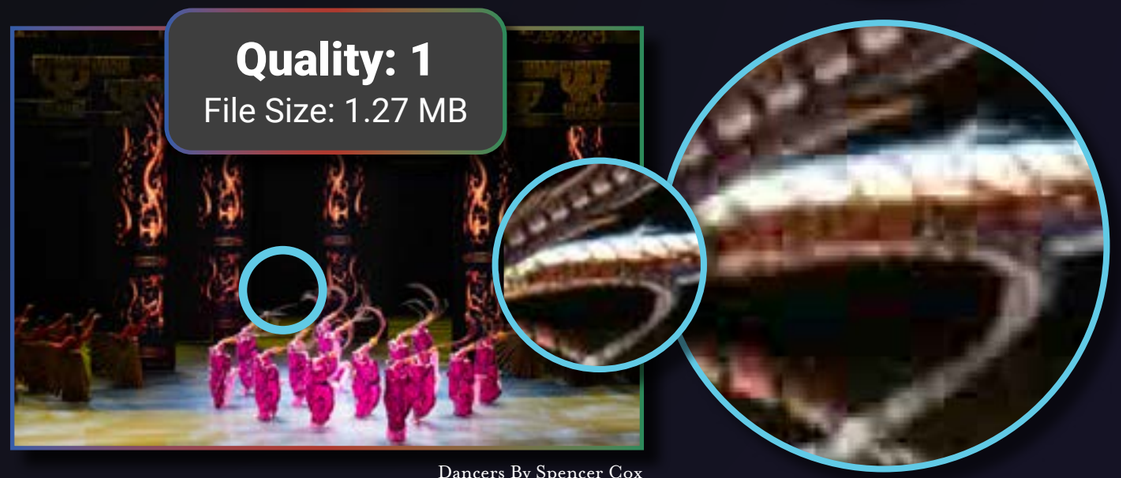




## Photoshop Steps:

When exporting images in Photoshop, users are given several different export methods depending on their workflow and output needs. For this demonstration, we focus specifically on the "Save a Copy" method using the JPEG format. As shown to the right, exporting with Save a Copy involves a simple sequence of steps. First, the user selects File and then presses Save a Copy. A dialog box then appears, allowing the user to choose the desired file format and adjust the image quality settings. Once these options are set, the image can be saved to the selected destination folder. This method provides precise control over compression and file size while maintaining visual quality.




Dancers By Spencer Cox

# JPEG: SAVE AS

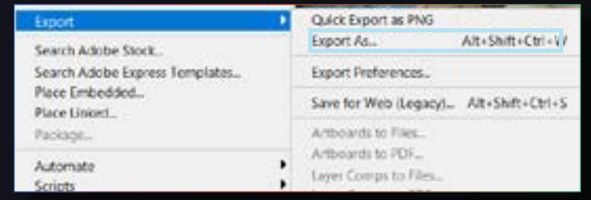
## Export_Save As (SA):

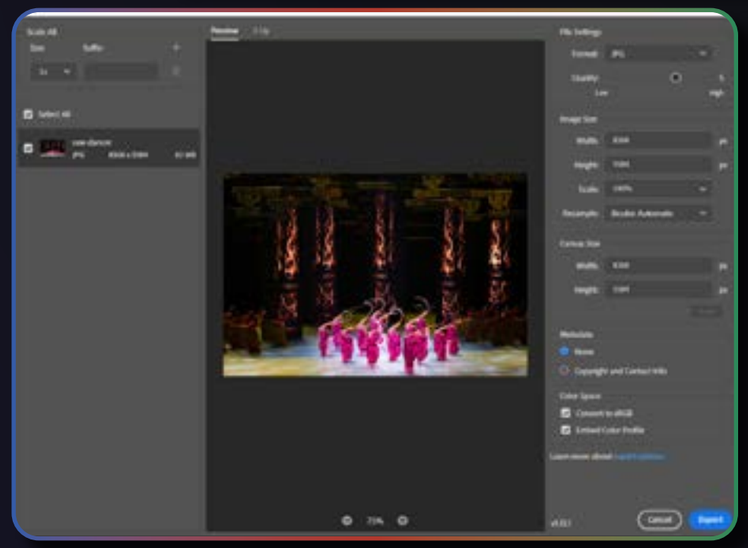
Similar to Save a Copy, reducing the quality from 7 to 4 produces minimal visible change in image appearance. At close range, pixelation remains nearly the same with only a small amount of additional artifacting, and from a distance, the image still appears crisp and detailed. The most significant difference is found in file size: at quality 7, the file measures 31.1 MB, while reducing it to quality 4 dramatically decreases the size to 3.12 MB, a much larger reduction than what was observed using Save a Copy.

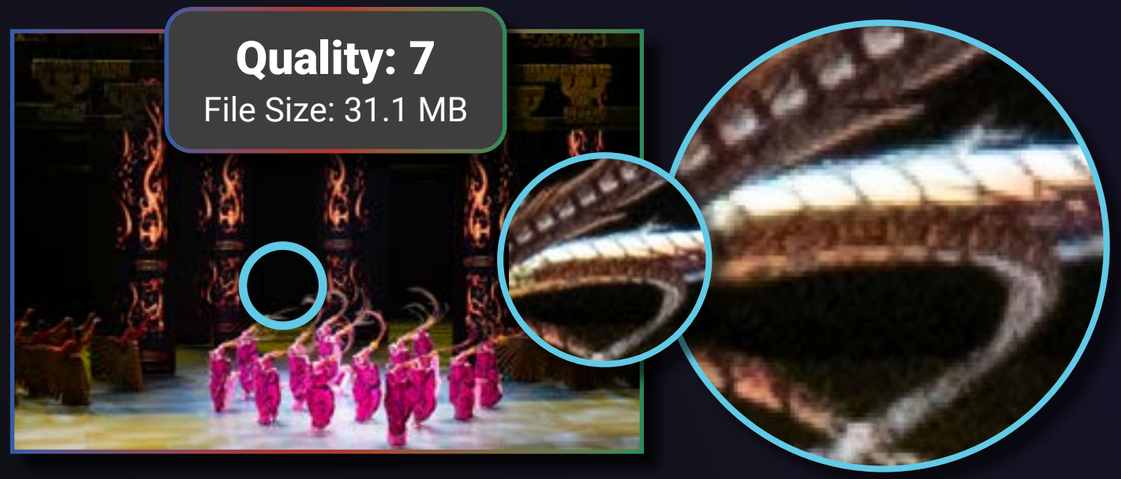
As expected, pixelation becomes increasingly apparent at quality 1, with heavy artifacting and visible blocking. While the file size at this level is extremely small, the severe loss in visual quality makes it unsuitable for professional design use. Although these artifacts are most noticeable in close-up views, the degradation affects the entire image overall. This demonstrates that while aggressive compression may reduce storage needs, it significantly compromises resolution and should be avoided when visual clarity is required.

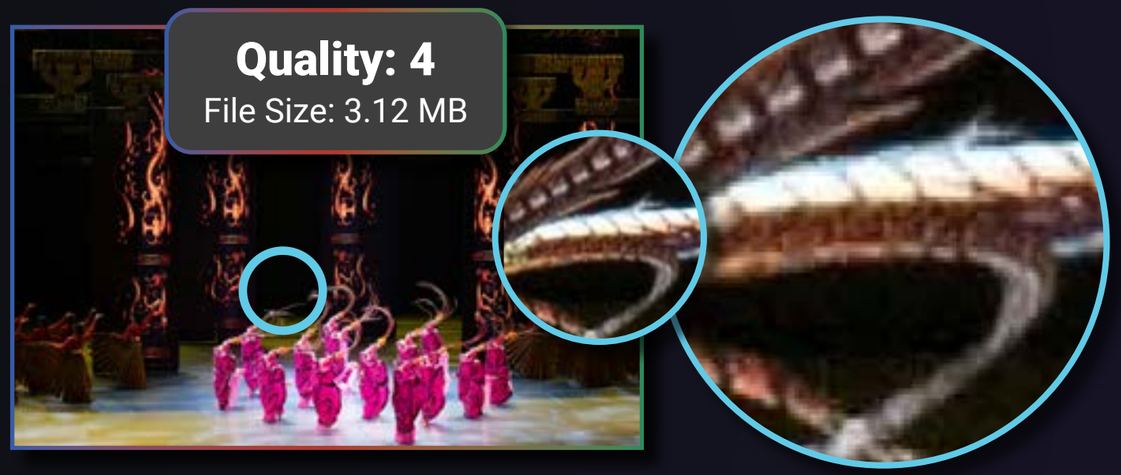
## Photoshop Steps:

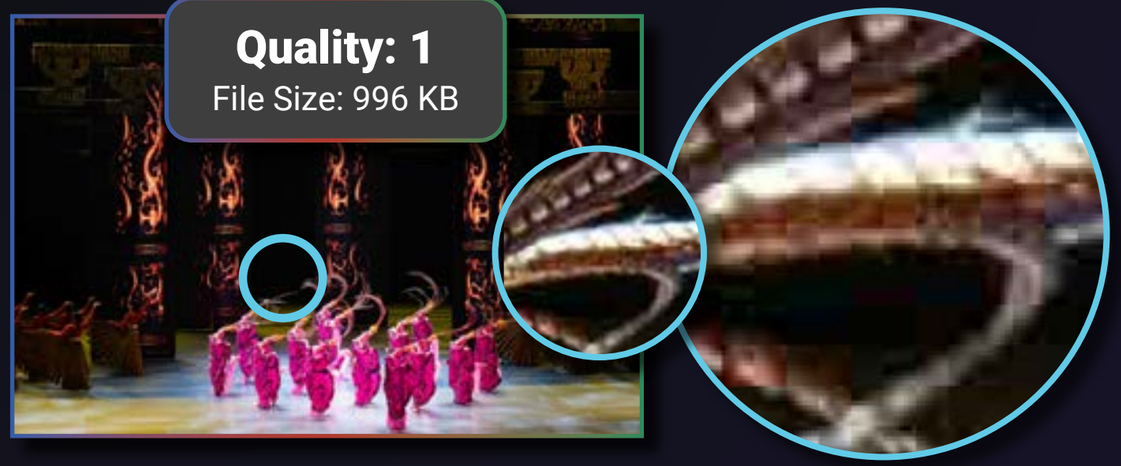
Just like exporting a Save As Copy, we still go to File, before hopping down to Export, and finally, choosing Export As. After this, we will see a dialogue box that will show us a preview of the image and give us the ability to adjust the sliders to find our preferred quality. There's a lot more information when it comes to the Export As dialogue box than the Save As Copy, which is extremely convenient when exporting photos.





# JPEG: SAVE FOR WEB

## Save For Web (SFW):

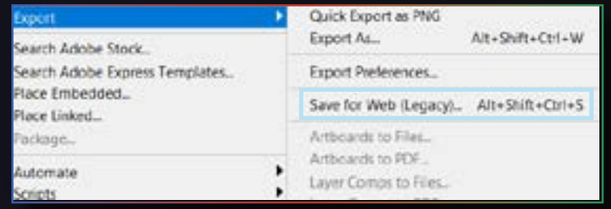
Similar to the previous export methods (Save a Copy and Export: Save As), there is little visible difference between images exported at quality 100 and quality 50. Both appear visually similar, with the most significant difference being file size, as the 100% quality image requires nearly five times more storage than the 50% version.

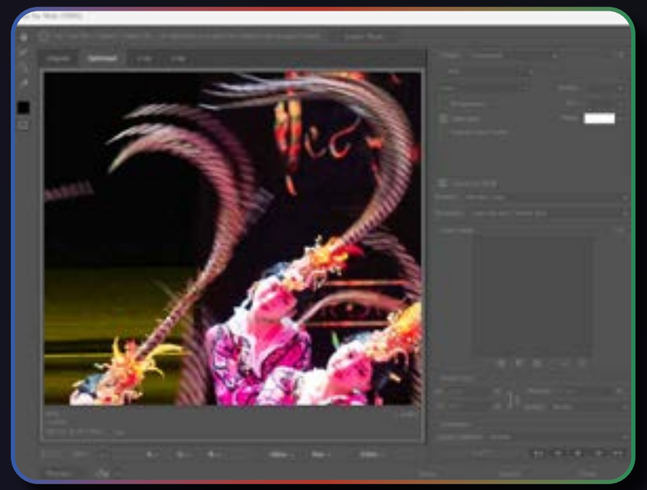
When comparing quality 100 to quality 1, visual degradation becomes more apparent, especially in the close-up views. Blocking and compression artifacts are clearly visible at the lowest quality, which reduces image clarity. Although the image may appear acceptable from a distance, up close it reveals the square pixel structure that causes the image to appear grainy and pixelated.

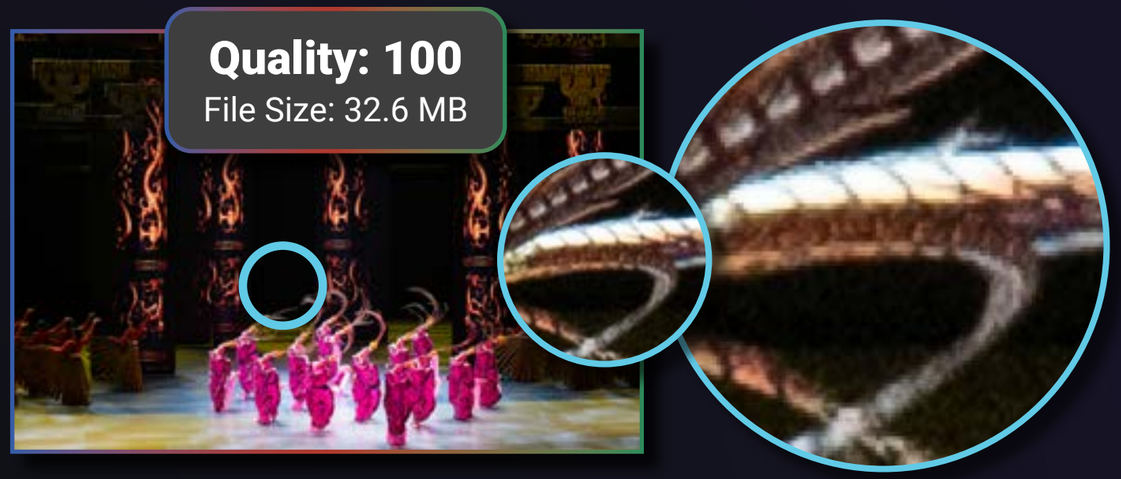
This comparison demonstrates how compression quality directly affects both file size and visual fidelity, helping designers determine the optimal balance between maintaining strong image quality and avoiding unnecessarily large file sizes.

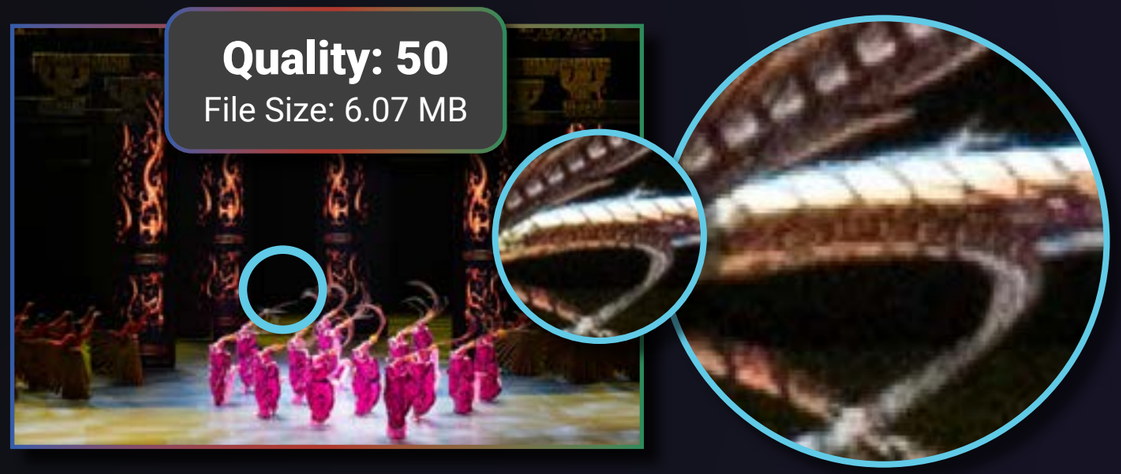
## Photoshop Steps:

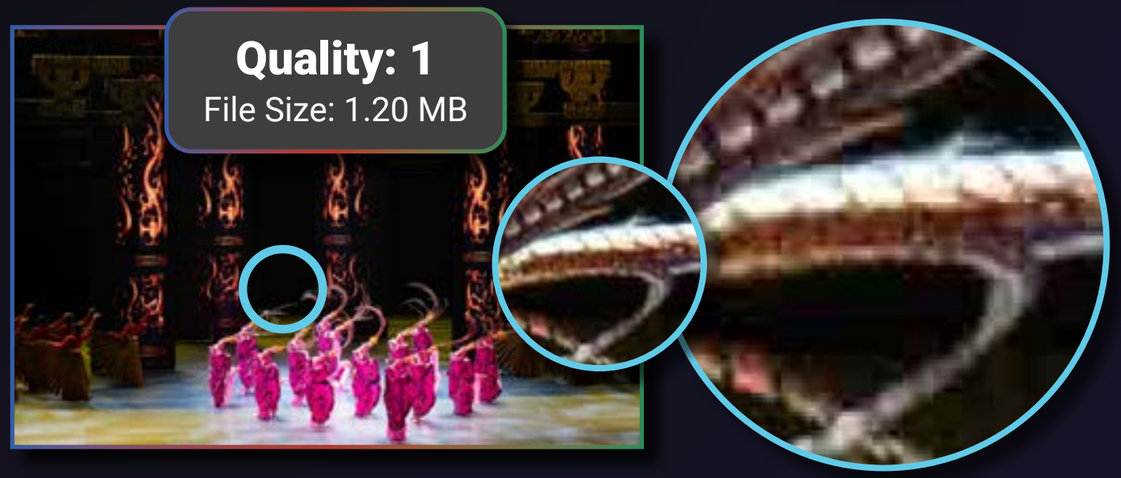
Lastly, as we explore the Save for Web, we get a dialogue box that appears after we push File, Export, and then Save for Web. Though very similar to the other dialogue boxes, this one is confusing, with a different layout and settings to become accustomed to. Out of the three different export options, Save a Copy, Save As, and Save for Web, the export for Save for Web was the most confusing to navigate and understand. Because of that, it's not my preferred method of exporting.





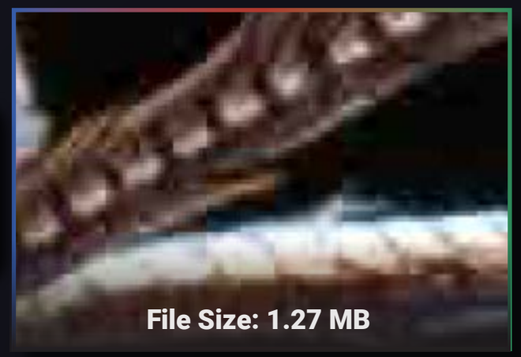
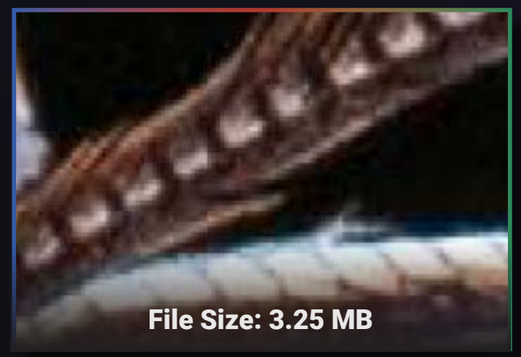
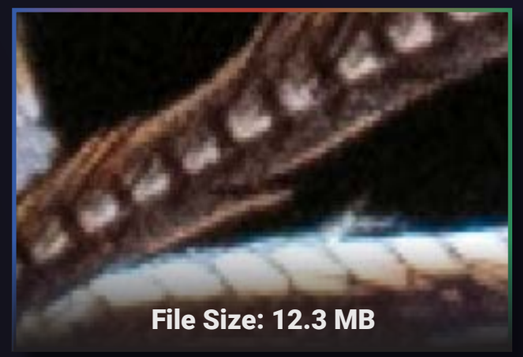
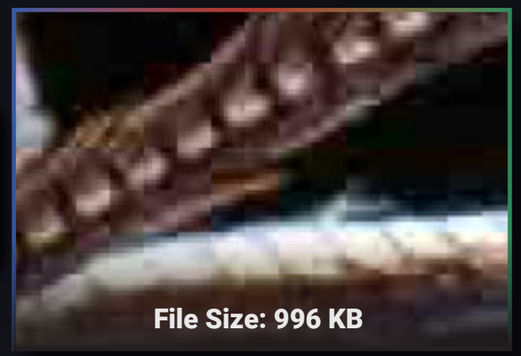
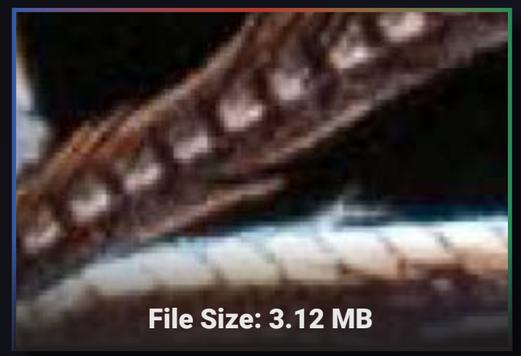
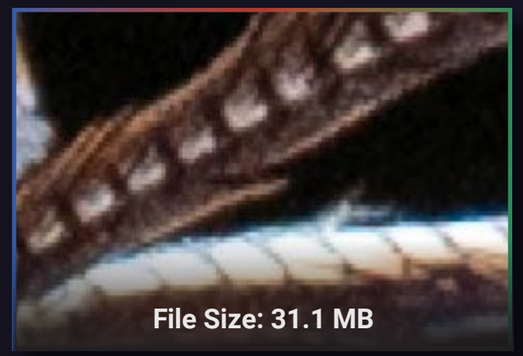
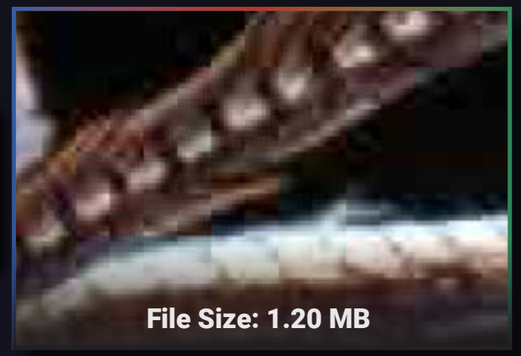
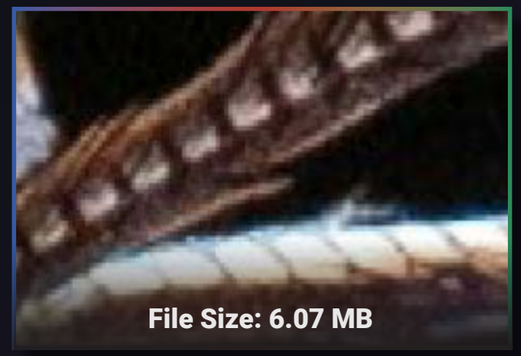
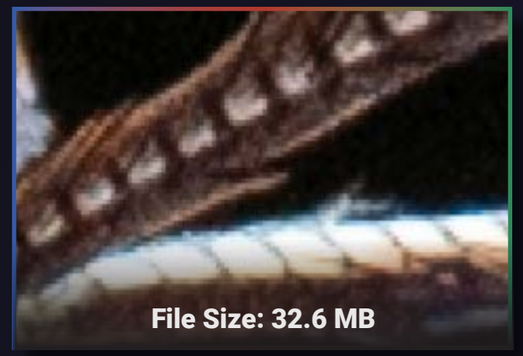
# JPEG: COMPARISON

## Side-By-Side

According to the images on the right, nine close-up examples display the different export quality options available in Photoshop. Above each column is the assigned quality level, and above each image is its corresponding file size. When examining the lowest-quality image, visible blocking and artifacting appear. While these distortions are not noticeable from a distance, they become clearly visible when viewed up close, revealing significant degradation in image detail.

As we move to the medium-quality images, there is a clear improvement. Blocking becomes far less prominent, artifacting is reduced, and overall sharpness increases. The individual pixel structure is smaller and more refined, allowing the image to retain much of its visual clarity even at close magnification. When comparing the medium- and high-quality images, the visual differences become minimal. The higher-quality version appears slightly less grainy, but overall, both images appear very similar in clarity.

When the file sizes are compared, however, the difference becomes significant. There is a dramatic reduction in file size between high and medium quality, despite only a minor change in visible resolution. This demonstrates that designers can significantly reduce file size while maintaining strong visual quality. Understanding this balance is crucial for optimizing images for web and digital use without sacrificing appearance.

| | Quality: Lowest | Quality: Medium | Quality: Highest |
|---|---|---|---|
| Save a Copy |  File Size: 1.27 MB |  File Size: 3.25 MB |  File Size: 12.3 MB |
| Export_ Save As |  File Size: 996 KB |  File Size: 3.12 MB |  File Size: 31.1 MB |
| Export_ SFW |  File Size: 1.20 MB |  File Size: 6.07 MB |  File Size: 32.6 MB |

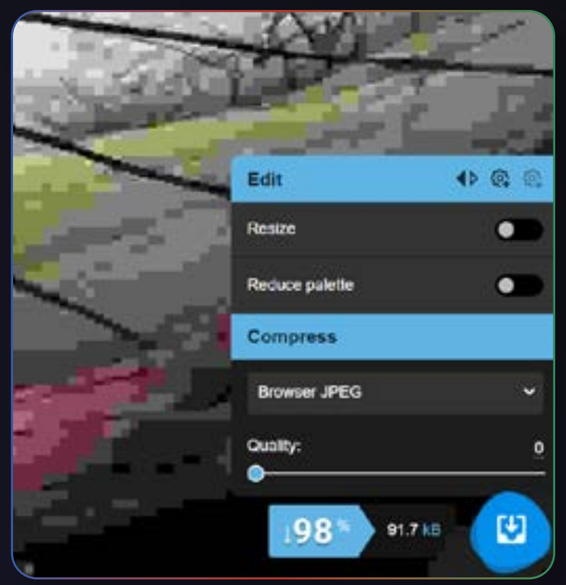
# SQUOOSH: JPEG

## JPEG Compression

Unlike Photoshop, Squoosh shows a noticeable decrease in image quality when moving from quality 100 to quality 50. At this level, pixels become visibly grainier, and compression artifacts begin to appear. When the quality is reduced further to quality 1, there is a significant loss of visual information—color becomes severely distorted, and the image becomes nearly indistinguishable.

While Squoosh is an effective tool for adjusting image output and testing compression visually, it struggles when processing large files measured in megabytes (MB) at high quality. When attempting to compress large files to lower quality settings, Squoosh often requires several minutes to process before the image can be saved. In contrast, Photoshop handles both quality adjustments and large file compression more efficiently, processing changes quickly while still producing a manageable file size. This highlights Photoshop's advantage in professional workflows that require frequent export adjustments and large file handling.

Below are two different JPEG compressions, but one has color and one is devoid of color. Depending on the different JPEG compressions, you will get different results, so it's important to know how the results will turn out as well as the file size.



Umbrella by Michael Harper

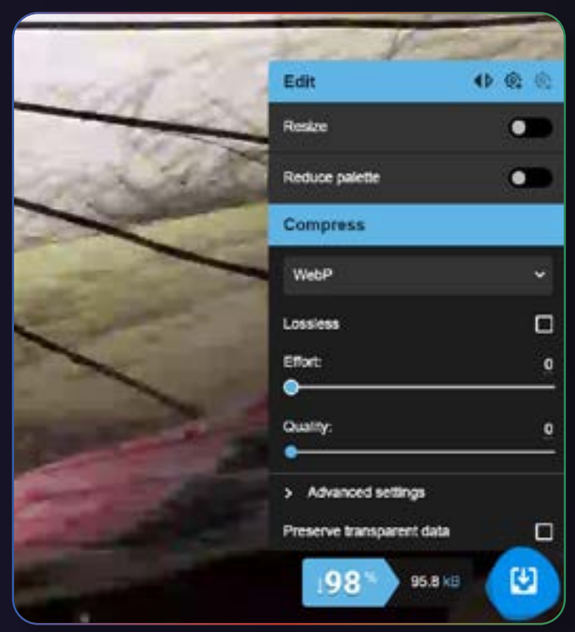
# SQUOOSH: WEBP

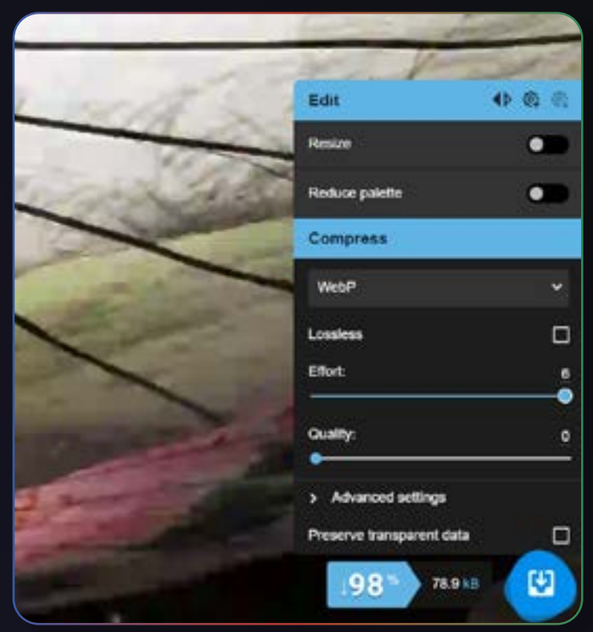
## WEBP Compression

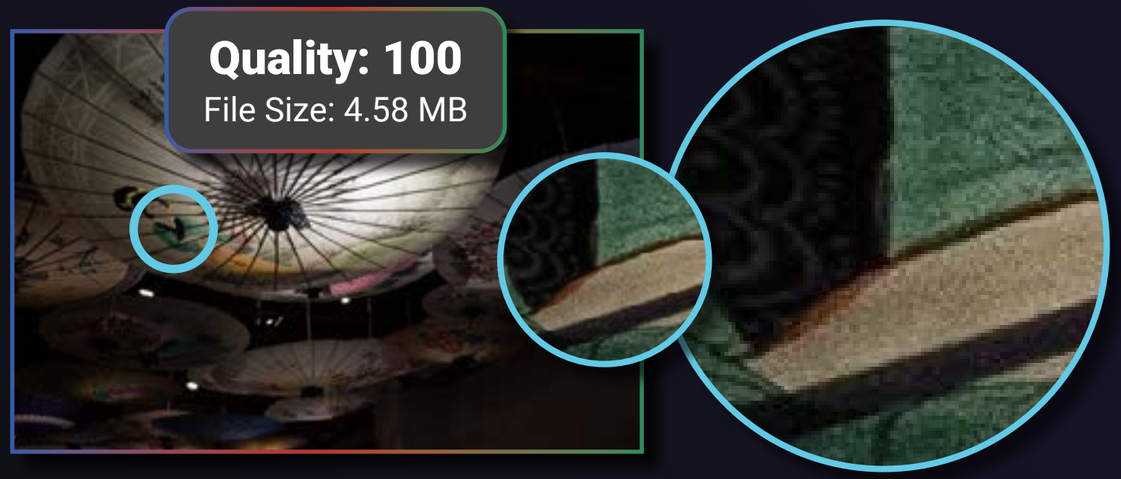
Unlike exporting a JPEG in Squoosh, exporting a WEBP file includes an additional setting called the Effort slider. Lower Effort values allow the file to compress more quickly, while higher values increase processing time to achieve slightly more optimized compression. However, when an image is set to lossless, the Effort level does not affect the final image quality, since no visual data is discarded during compression.

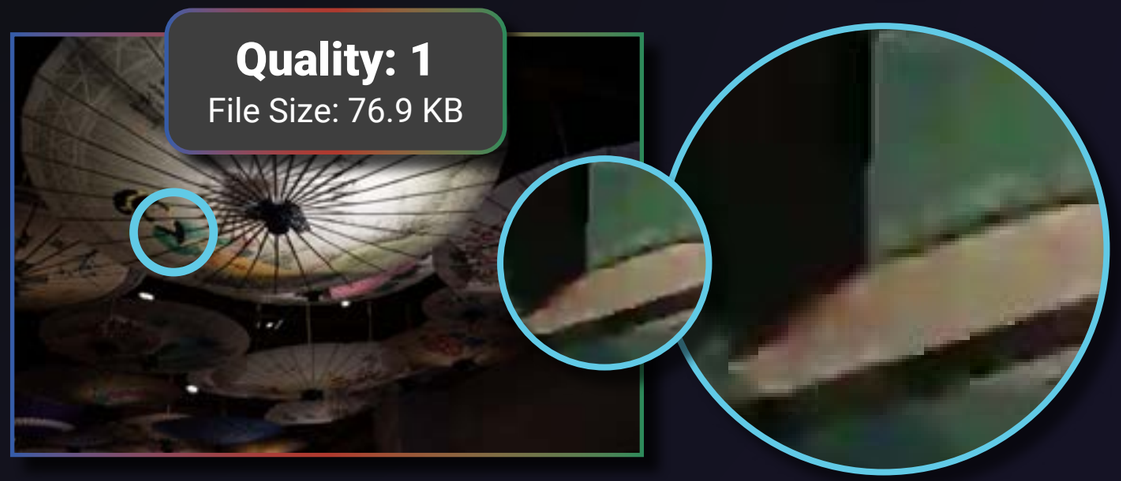
Similar to JPEG compression, a WEBP file exported at quality 100 still results in a file size measured in megabytes, though it is significantly smaller—approximately one-fourth the size of the original 12.4 MB file. This demonstrates WEBP's ability to preserve high visual quality while achieving substantial file-size reduction.

Below are two images. The only difference is the effort scale. On the left, the effort is set to 0 while the effort on the right is set to 6. With more effort, the image is smoother, but with less effort, the image is pixelated and choppy. It doesn't appear to be as smooth as the image with more effort.





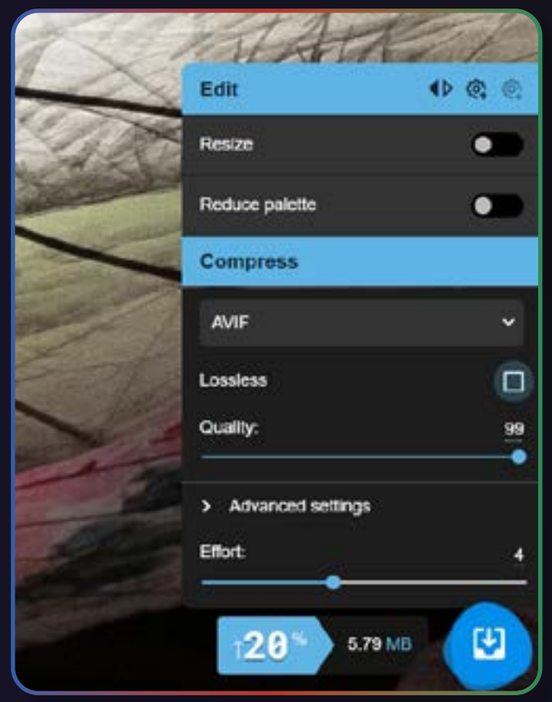
# SQUOOSH: AVIF

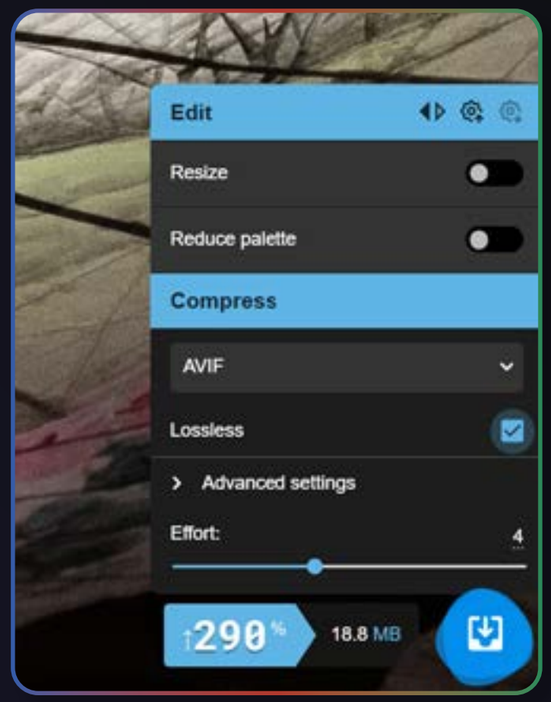
## AVIF Compression

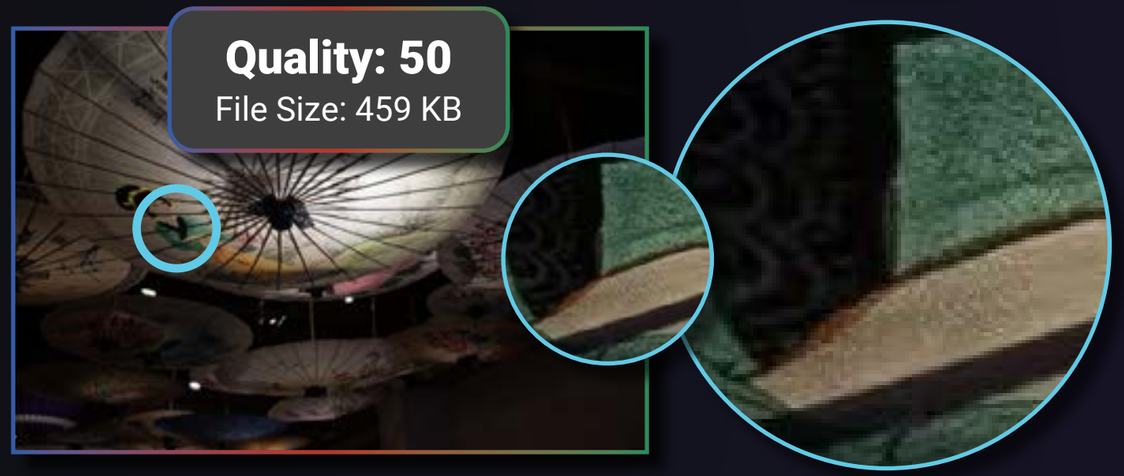
Similar to JPEG and WEBP compression, the AVIF image file shows very little visible difference between quality 100 and quality 50. These two versions appear nearly identical, while JPEG and WEBP display slightly more blocking and artifacting at comparable settings. However, when the AVIF export is reduced to quality 1, the image becomes noticeably blurred, as if viewed from a distance or through a hazy surface. At this level, fine detail is significantly lost, clearly demonstrating the effects of aggressive compression.

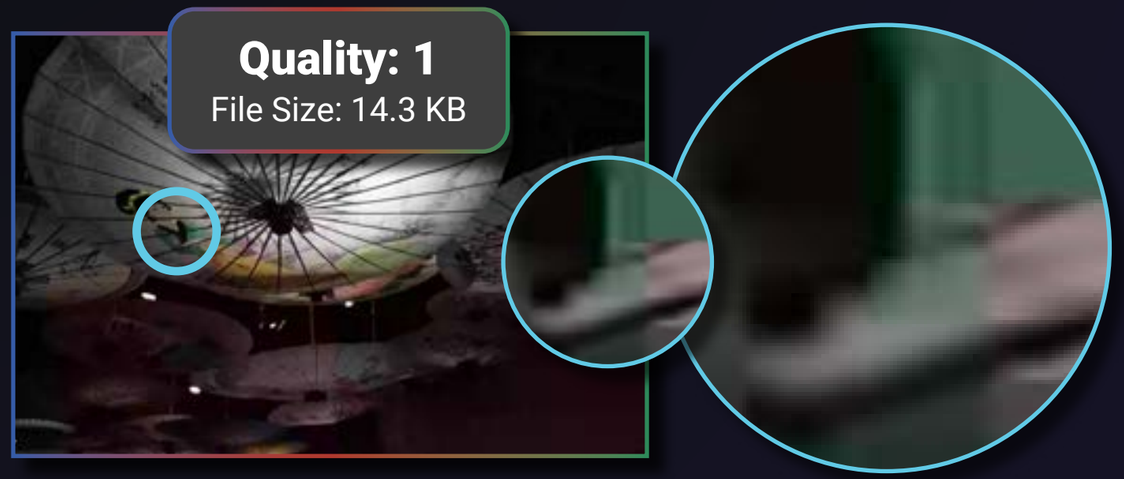
This information is especially valuable for designers exporting images through Squoosh for client websites. Understanding where compression begins to compromise visual clarity helps designers make informed decisions when Photoshop is not available and ensures that images remain optimized without exceeding acceptable resolution loss.

Unlike exporting a JPEG in Squoosh, exporting a WEBP file includes an additional setting known as the Effort slider. Lower Effort values allow the file to compress more quickly, while higher values require longer processing time but may result in slightly more optimized compression. However, when the image is set to lossless, the Effort level does not affect final image quality, since no visual data is discarded during compression.





# SQUOOSH: COMPARISON

## Side-By-Side

When viewed side by side at close range, a clear degradation in resolution becomes visible when comparing high- and low-quality exports. At the lowest quality, JPEG shows the most severe loss of visual information, including noticeable color degradation. In contrast, WEBP and AVIF retain their overall color, but much of the image definition is lost due to heavy blocking and compression artifacts.

As we move from low to medium quality, there is an immediate improvement in resolution. Artifacting is still present but occurs less frequently and is significantly less pronounced. At this level, JPEG and WEBP appear fairly similar in overall quality, while AVIF retains slightly more fine detail, particularly in the green regions of the image. Across all formats, some loss of dark tonal values is observed at medium quality, causing portions of the image to appear slightly brighter than at high quality. While the difference between medium and high quality is subtle when viewing the image as a whole, side-by-side comparison reveals mild artifacting at the medium level.

Another noticeable change is in color behavior as compression increases. At lower quality levels, AVIF shifts toward a cooler color tone, while WEBP maintains warmer color consistency. This demonstrates that each compression format not only affects sharpness and detail but also influences overall color rendering.

# PHOTOSHOP: WEBP

## WEBP Compression

Among JPEG, WEBP, and AVIF, WEBP demonstrates the strongest overall compression performance in this comparison. It maintains the clearest and most consistent image quality across multiple compression levels. Using a single WEBP image, a chip chart was created to examine how visual quality changes as the image is scaled and compressed.

Based on these results, noticeable grain and loss of detail do not become prominent until approximately quality 40, at which point visual information begins to break down. This comparison highlights the efficiency of WEBP as a compression format, making it a valuable option for designers who need to balance high visual quality with low file size when preparing images for web use.

Taxi By Spencer Cox

Compression Quality →

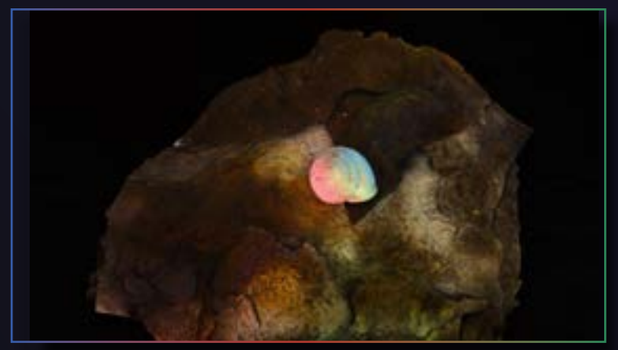
| | Quality: 1<br>File Size: 312 KB | Quality: 20<br>File Size: 1.14 MB | Quality: 40<br>File Size: 2.40 MB | Quality: 60<br>File Size: 3.89 MB | Quality: 100<br>File Size: 23.3 MB |
|---|---|---|---|---|---|
| 100% | | | | | |
| 300% | | | | | |
| 500% | | | | | |

Zoom Level ↑

# PHOTOSHOP: FORMAT COMPARISON

## Compressions in Photoshop

In this chip chart, we can see a comparison of various image formats, their quality settings, and their corresponding file sizes. Based on this data, TIFF produces the largest file size, while AVIF achieves the smallest. Overall, even though TIFF, PNG, and AVIF files are larger at 50% quality, they still maintain a similar level of sharpness and clarity as JPEG and WEBP images.

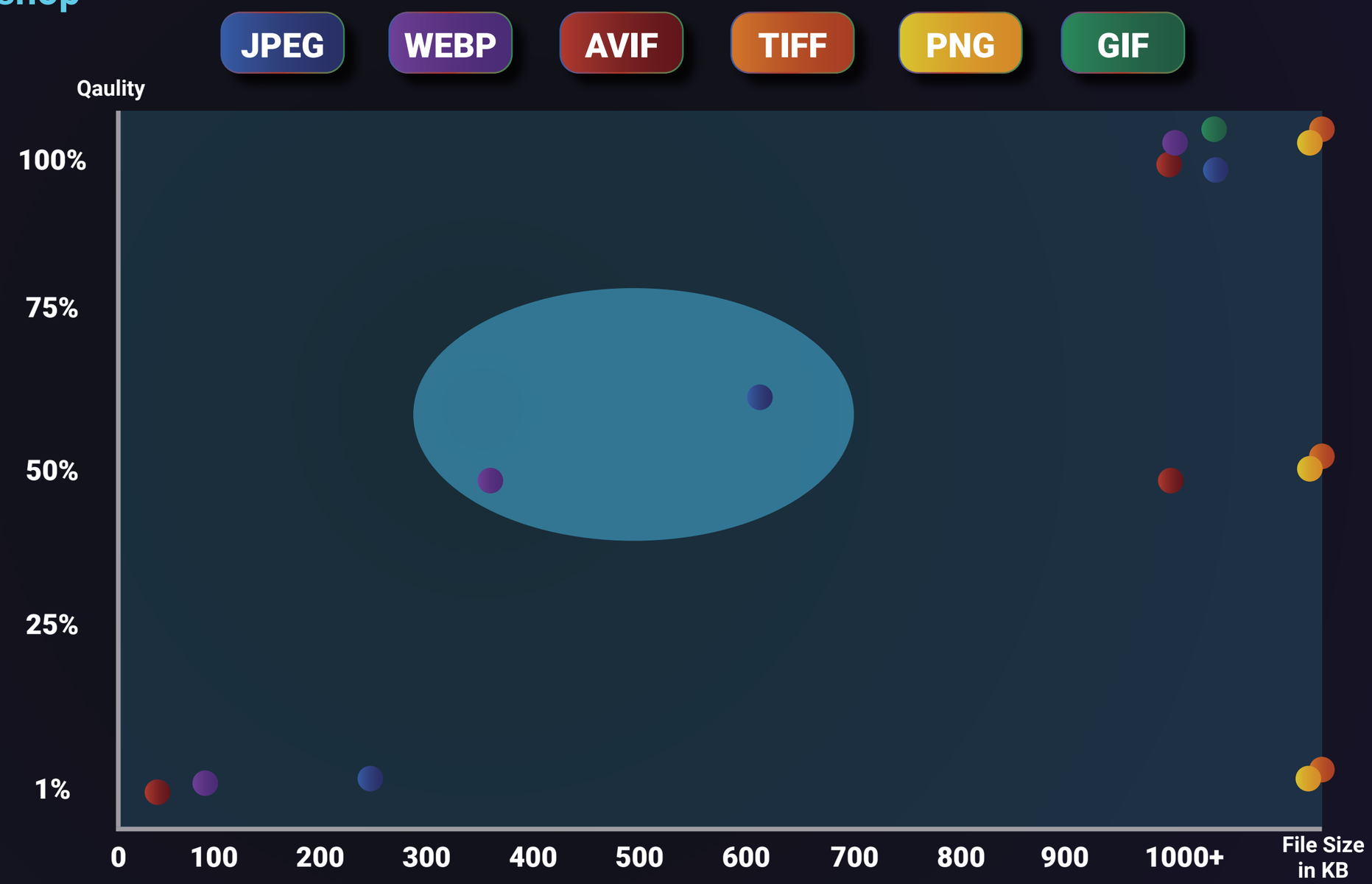


# PHOTOSHOP: FORMAT COMPARISON

## Compressions in Photoshop

To the right, the graph illustrates the relationship between image quality and file size across different compression levels. Images saved at maximum quality retain the most detail and remain crisp, but their file sizes often reach into the megabyte range. In contrast, images saved at minimum quality have much smaller file sizes but appear blurry and pixelated. At around 50% quality, however, the file size stays reasonable while the image remains sharp and clear, free of noticeable pixelation or blurriness.

Based on the information about various image formats and their quality levels, we can conclude that even at 50% quality, images remain crisp and clear. It's only when the quality drops to its lowest setting that the images become noticeably blurry and pixelated.



# 03 Audio

# Formats

## Overview

Audio files contain large amounts of data, making compression essential for streaming and digital storage. Audio codecs control how sound is compressed, how much information is preserved, and how accurately audio is reproduced on different devices. The distinction between lossy and lossless formats plays a major role in sound quality, file size, and streaming efficiency. In this section, we will explore MP3, AAC, WAV, and FLAC formats and compare how platforms such as Spotify, Apple Music, YouTube Music, and Tidal handle audio compression and playback quality.

# AUDIO FORMATS

There are many different containers for audio, but for now, we are going to address, MP3, FLAC, WAV, and AAC. Later, we will go more in depth.

## .MP3 (.mpga & mpeg3)

**Stands for:** Moving Picture Experts Group – Audio Layer III
**Created:** late 1980s–early 1990s
**Compression:** Lossy
**History:** MP3 is maintained and standardized by the International Organization for Standardization and the International Electrotechnical Commission. It was developed in 1898 by Karlheinz Brandenburg, Ernst Eberlein, Heinz Gerhouser, and others at Fraunhofer IIS.
**Facts:**

- MP3 uses bitrate — The bitrate determines how much data is used per second of audio
- The maximum bitrate for MP3 is 320 kbps (kilobits per second)
- MP3 can also use Variable Bitrate (VBR), meaning it adjusts the bitrate depending on the audio complexity
- It's a perceptual encoding format —
- High audio fidelity due to lossless, uncompressed format
- Supports multiple audio channels (mono, stereo, and beyond)
- Compatible with most operating systems and audio editing software
- Simple, extensible format that can include metadata

## .FLAC

**Stands for:** Free Lossless Audio Codec
**Created:** 2000
**Compression:** Lossless
**History:** The format was developed by the Xiph.Org Foundation and is open-source. It was later formalised and published as an Internet standard under Internet Engineering Task Force (IETF) as RFC 9639.
**Facts:**

- Designed specifically for audio
- Can compresses audio without removing any data
- Digital audio compressed by FLAC's algorithm can typically be reduced to between 50 and 70 percent of its original size and decompresses to an identical copy of the original audio data.
- It's the preferred archival format for many libraries, audio engineers, and streaming platforms that handle high-resolution content (like Tidal, Qobuz).
- Supported across major platforms (Windows, macOS, Linux, Android) and most modern players.

## .WAV (.wav, .wave)

**Stands for:** Waveform Audio File Format
**Created:** 1991
**Compression:** WAV (ADPCM or other codec) Lossy
**Compression:** WAV (PCM) Lossless/uncompressed
**History:** The format was developed jointly by Microsoft and IBM and is based on the Resource Interchange File Format (RIFF) container.
**Facts:**

- WAV is a subset of RIFF (Resource Interchange File Format) — a chunk-based container format
- Typically used for uncompressed Linear PCM (Pulse-Code Modulation) audio.
- WAV files store raw, uncompressed audio data in the data chunk using PCM encoding, meaning the audio is represented as a series of samples
- meaning it reconstructs the waveform from compressed data using psychoacoustic models, so the "bit depth" is approximate and not fixed.
- When MP3's file is too small, small details will be noticed like the room reverb, background instruments, and stereo width get flattened

## .AAC

**Stands for:** Advanced Audio Coding
**Created:** 1997
**Compression:** Lossy
**History:** The codec is standardized by Moving Picture Experts Group (MPEG) under the umbrella of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC).
**Facts:**

- .m4a is an audio containers that uses AAC
- .mp4, .m4b, .m4p, .3gp are containers that might use AAC
- AAC was developed as a successor to MP3, designed to improve audio quality at similar bit-rates and support more channels and sampling rates.
- The first standard version was introduced in 1997 under MPEG-2 Part 7PNG is an open format and can be edited in any program
- AAC audio can be carried in .3gp files since they are derived from the MPEG-4 Part-12
- When the bitrate is low, it removes parts of the sound but handles it better than MP3

## .ALAC

**Stands for:** Apple Lossless Audio Codec
**Created:** 2004
**Compression:** Lossless
**History:** IETF published the open technical specification.
**Facts:**

- ALAC is a codec, not a container, so it usually appears inside these containers: .m4a, .mp4, .caf, .mov
- ALAC supports 16-bit, 24-bit, and higher
- ALAC remains proprietary, limited mainly to: iTunes, iPods, iPhones, and Apple Software
- Fully integrated into: Apple Music, macOS, iOS, Final Cut Pro, Logic Pro, Royalty-free and open-source (since 2011)
- Competes directly with: FLAC, WAV, AIFF

# AUDIO: INTRO

## Intro

With so many music-streaming platforms available today, audio quality can vary more than people realize. To understand how each service truly sounds, I compared major platforms using different types of listening devices.

The device you use plays a huge role in determining how good your music actually sounds. Whether someone listens through Bluetooth or a wired DAC (Digital-to-Analog Converter) can dramatically change the overall audio experience. Each method handles sound differently, which affects clarity, depth, and detail.

For this comparison, I used three listening setups: Sony WH-1000XM4 Wireless Noise-Cancelling Headphones and Apple AirPods (2nd Generation). Each one shapes the sound in its own way, revealing how equipment and connection quality influence what we hear.

**Device used for test:**
iPhone 13 (Blue Tooth sharing)

**Listening Experience Comparison: Airpods vs. Sony WH-1000XM4**

**Sony WH-1000XM4**
*Wireless Noise-Cancelling*

- **Best sound quality**
- **Great bass**
- **Excellent ANC**

**Perfect for lossless**

**AirPods**
*(1st/2nd Gen)*

- **Clear vocals**
- **Weak bass**
- **No noise cancelling**

**Not great for lossless/hi-res**

# AUDIO: STREAMING

## Spotify

Spotify is one of the most well-known music platforms today, serving approximately 713 million active listeners. It offers a huge library of music and gives people the option to listen for free or upgrade to Spotify Premium for $11.99 per month. While most listeners use the free version, around 281 million subscribers pay for Premium, which includes Spotify's new lossless 24-bit/44.1 kHz FLAC quality.

Spotify Screenshot

Spotify Screenshot

To test these differences, I used Sony WH-1000XM4 Wireless Noise-Cancelling Headphones, which are well known for their premium ANC and sound purity. On Spotify's free version, there was noticeably more background noise. The music sounded "grainier," and the instrumental layers felt scratchy and distant—almost as if the sound were passing through a vibration-resonance speaker. Compared with other platforms, Spotify Free created a barrier between me and the music, making it feel less clear and less immersive.

Spotify Premium was a major improvement, offering a clean and detailed sound. However, at higher volumes, I still noticed a slight static or background "hiss" that didn't appear as strongly when listening through Apple Music or Tidal. The background elements in the mix felt more pronounced on Spotify, sometimes overpowering the main vocals.

## Apple Music

Apple Music is smaller than Spotify in subscriber count, with around 94 million paid listeners, but it offers extremely high-quality audio. Every song displays a small badge that shows its format, such as:

- Lossless (16-bit / 44.1 kHz ALAC)
- High-Res Lossless (24-bit / 48–192 kHz ALAC)
- Dolby Atmos (Spatial Audio)

Apple Music consistently delivered fuller, warmer sound. Vocals felt richer and deeper, while instruments blended more smoothly. Nothing competed for attention—each part of the track had a clear role. The only downside was that occasionally, the bass felt slightly "soft" or less crisp compared to Tidal. But overall, Apple Music created a clean, pleasant, and balanced listening experience.

Apple Music Screenshot

Tital Screenshot

## Tidal

Tidal is less widely used than Apple Music, but it has the same base subscription price of $10.99 per month. Tidal provides CD-quality Lossless (16-bit/44.1 kHz FLAC) and Hi-Res FLAC up to 24-bit/192 kHz.

Sound-wise, Tidal was extremely similar to Apple Music, but slightly louder and more dynamic. Tidal did a better job of highlighting individual elements in a song. When the drum section became prominent, you could feel it more clearly than on Apple Music. While Apple Music can sometimes feel "flat," Tidal feels more textured and expressive. Another advantage is that Tidal clearly labels each track's format, which audiophiles appreciate. However, for casual listeners, this isn't a major selling point and may not justify the monthly cost.

YouTube Music Screenshot

## YouTube Music

YouTube Music offers free listening or an upgraded ad-free experience. However, its maximum quality is 256 kbps AAC, with no lossless options. Compared to other platforms, YouTube Music sounded noticeably louder in every aspect—vocals, instruments, and bass all felt boosted. High-pitched sounds and “S” consonants were especially sharp. While these flaws were less noticeable through car speakers, they were very obvious when using high-quality headphones like the Sony WH-1000XM4.

When listening through Apple’s AirPods, the music felt like it was coming from the front of the earpiece, while music on other platforms sounded as though it was coming from behind.

## Pandora

Pandora has existed for many years and had roughly 46 million active listeners in 2023. Its popularity has decreased over time, and its maximum streaming quality tops out at 192 kbps AAC—significantly lower than Tidal, Spotify Premium, or Apple Music.

Using the Sony WH-1000XM4, the difference between Pandora and Apple Music was subtle at first. But switching to Apple AirPods (1st generation) made the differences clearer. Pandora sounded muffled, fuzzy, and less crisp overall. The volume also felt inconsistent, with some parts of songs sounding almost “echoed” or unbalanced.

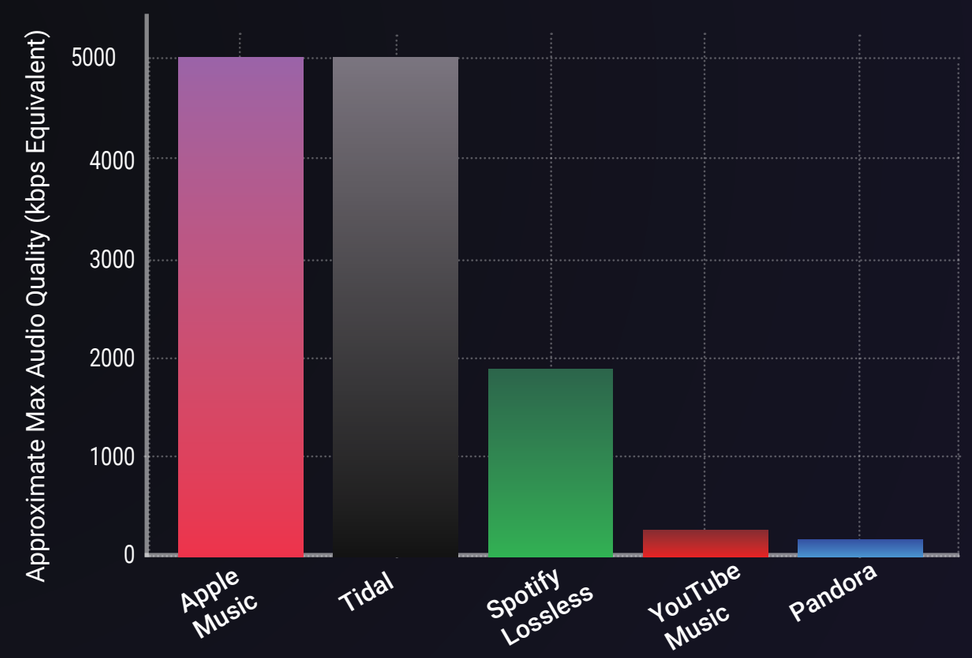
Pandora Screenshot

## Conculsion

Ultimately, Apple Music and Tidal provided the most consistent high-quality sound. Spotify Premium offered good quality but still fell short of lossless competitors, while Spotify Free, YouTube Music, and Pandora showed clear limitations when tested with high-quality headphones.

### Comparison of Max Audio Quality Across Streaming Services

| Service | Max Quality / Resolution | Notes & Format Info |
|---|---|---|
| Apple Music | Up to 24-bit / 192 kHz Hi-Res Lossless | Uses ALAC; supports true lossless & hi-res. |
| Tidal | Up to 24-bit / 192 kHz Hi-Res FLAC | HiFi+ tier offers highest fidelity. |
| Spotify | Up to 24-bit / 44.1 kHz FLAC | Lossless tier recently introduced |
| YouTube Music | Up to 256 kbps AAC / OPUS | No lossless; premium “High” setting is 256 kbps. |
| Pandora | Up to 192 kbps (paid tier) | Uses lossy formats; lower than lossless services. |



# 04 Video Formats

## Overview

Video formats play a critical role in how motion media is stored, transmitted, and displayed across different devices and platforms. Behind every video file is a codec that determines how the video is compressed, how much data it uses, and how well visual quality is preserved. Choosing the right video format affects file size, playback compatibility, streaming efficiency, and overall image quality. In this section, we will examine common video formats and codecs—including MPEG, HEVC, AV1, and MOV—and compare how they are used across streaming platforms such as Netflix, Amazon Video, and Disney+ using Sony's The Amazing Spider-Man as a test example.

# VIDEO FORMATS

When it comes to video formats, there are several different types of CODECS incluidng MPEG, HEVC, AV1, and MoV.

## MPEG (.mpg, .mpeg, .mpe)

**Stands for:** Moving Picture Experts Group
**Created:** 1988
**Compression:** Lossy, but can be Lossless
**History: Joint working group of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), formally ISO/IEC JTC 1/SC 29/WG 11.** It produced major standards such as MPEG-1 (1993) and successive families (MPEG-2, MPEG-4, etc.).
**Facts:**

- MPEG-1 —digital video + MP3
- MPEG-2 — DVDs + broadcast TV
- MPEG-4 — streaming, video chat, animations
- MPEG-7 — multimedia metadata
- MPEG-21 — digital media frameworks
- MPEG-1 and MPEG-2 video compression use the same kind of perceptual psychology as MP3
- Some parts of MPEG include lossless modes
- Used for DVD video, broadcast TV, and satellite TV
- MPEG-4 ALS (Audio Lossless Coding) — lossless audio

## .HEVC

**Stands for:** High Efficiency Video Coding
**Created:** 2013
**Compression:** Lossy, but can be Lossless
**History:** Together, Moving Picture Experts Group and Video Coding Experts Group form the Joint Collaborative Team on Video Coding (JCT-VC), the body that created HEVC.
**Facts:**

- HEVC is used by Netflix, Prime Video, Apple, UHD Blue-ray, and iPhones.
- Lossless HEVC kills the compression benefit and it's not universally supported by hardware decoders
- H.264 couldn't compress large-resolution video efficiently anymore, so HEVC was created to reduce file sizes dramatically while keeping high quality.
- HEVC supports: HDR (High Dynamic Range), Wide color gamut (Rec. 2020), High bit depths (10-bit, 12-bit), and High frame rates (up to 240fps)

## .AV1

**Stands for:** AOMedia Video 1
**Created:** 2018
**Compression:** Lossy
**History:** It was created in 2018 by the Alliance for Open Media (Google, Netflix, Amazon, Microsoft, Apple, Meta, etc.)
**Facts:**

- It is royalty-free, unlike HEVC
- It competes with HEVC (H.265) and is used for: Youtube, Netflix, Chrome, Firefox, Edge, as well as some TVs and phones
- AV1 (AOMedia Video 1) was designed as a high-efficiency lossy compression format, meaning it reduces file size by removing visual information humans are unlikely to notice.
- AV1 was designed to replace HEVC
- AV1 is 20–30% better compression than HEVC (H.265)

## .MoV (.QTFF)

**Stands for:** QuickTime Movie Format
**Created:** 1991
**Compression:** Container
**History:** MoV was developed by Apple and partially standardized under ISO/IEC (because QTFF influenced the MP4 spec)
**Facts:**

- MOV is the file extension used for Apple's QuickTime Movie format.
- MOV is based on the QuickTime File Format (QTFF), which later became the foundation for the ISO Base Media File Format (ISO/IEC 14496-12) — the same structure used by MP4.
- MOV is a container, not a codec, and can hold H.264 video, HEVC/h.265 Video, ProRes video, AIC, AAC aduio, ALAC audio, MP3 audio, Timecode tracks, and subtitles.
- MOV is still widely used for ProRes, HDR video, and high-quality editing workflows.

# VIDEO: INTRO

Pioneer TV

**TV Model:** Pioneer PDP-5010FD

**Year:** 2008

**Type:** Plasma

**Resolution:** 1920 × 1080

**HDR:** No

**Impact on Study:** Naturally smooths & hides compression artifacts

Levono Laptop

**Laptop Model:** Lenovo 82XF

**Year:** ~2022–2023

**Type:** LCD / LED

**Resolution:** 1920 × 1200

**HDR:** No

**Impact on Study:** Reveals compression artifacts clearly due to sharp pixel structure and lack of natural smoothing

## Intoduction

For the video format section, I analyze the differences between a Pioneer plasma television and a Lenovo laptop display. These two screens represent very different viewing platforms that process and display compressed video in distinct ways. Because neither platform supports native 4K or HDR playback, both displays rely on standard dynamic range (SDR) and downscaling, which affects how compression artifacts appear.

## Pioneer TV

The Goonies, 1985

Despite being manufactured in 2008, the Pioneer PDP-5010FD plasma television was one of the highest-quality consumer televisions of its time. Plasma display technology naturally smooths motion and masks certain digital compression artifacts, such as blocking and banding. However, this television was designed for Full HD (1920 × 1080) viewing and was not built to handle native 4K content. In the image shown at the top, a still from The Goonies featuring Sean Astin is displayed on the Pioneer plasma TV. Although this is an older movie, the image demonstrates visible softness, shadow blocking, and mild artifacting, which are further influenced by the television's natural smoothing characteristics.

Modern films are often captured and processed using digital noise reduction and are mastered specifically for HD, 4K, and HDR delivery. In contrast, movies produced in the early 2000s and earlier were not originally mastered for high-resolution digital viewing, which contributes to their softer appearance and increased susceptibility to compression artifacts when streamed today.

## Levono Laptop

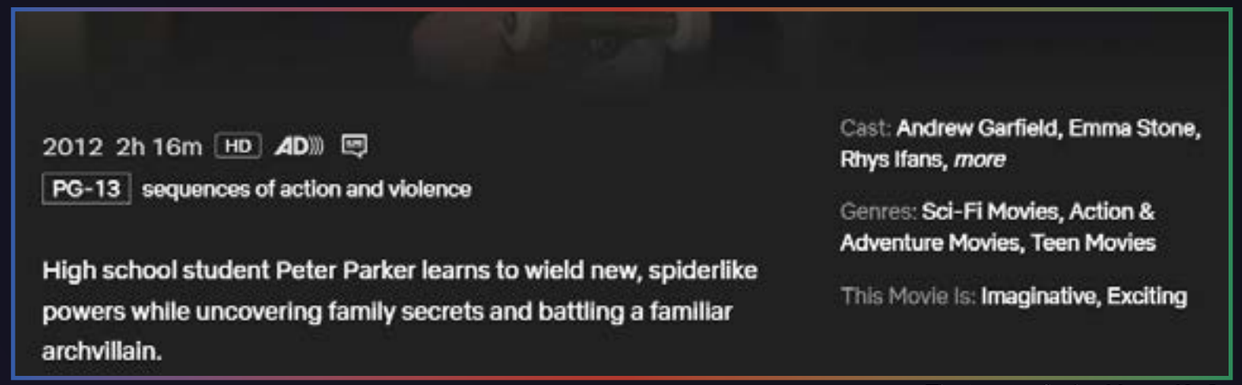
The image at the bottom is a direct digital screenshot taken from the Lenovo laptop. This image appears significantly darker than the photograph of the television because it captures the exact encoded pixel values without added camera exposure adjustments. While blocking and artifacting are slightly less visible in the screenshot, the image still lacks the sharpness and clarity associated with modern high-resolution, well-compressed sources.

Now that the two display platforms have been established, the analysis will move into The Amazing Spider-Man to compare streaming quality differences between Disney+, Amazon Prime Video, and Netflix.

# VIDEO: STREAMING

The Amazing Spider-Man on Disney+

The Amazing Spider-Man on Amazon Prime Video




The Amazing Spider-Man on Netflix

## Streaming Platforms

Each streaming platform has its own method for delivering movies. While some platforms operate in similar ways, others differ significantly. The purpose of this test is to identify and understand those differences.

When examining Amazon Prime Video, we notice a long list of icons that communicate technical and accessibility information about the selected movie. Amazon Prime Video displays many more icons than Netflix and Disney+. This is because Prime Video functions not only as a streaming service, but also as a rental store, a digital marketplace, and an accessibility hub within a single interface. In contrast, Netflix and Disney+ are considered “closed ecosystem” platforms and expose far fewer technical labels to their users.

Understanding these interface differences is important for designers because they influence how users perceive quality, accessibility, and technical transparency across platforms.

Below are the definitions for what each icon represents.

**Video Quality**

- HDR: High Dynamic Range
- UHD: 4K resolution
- HD: High Definition

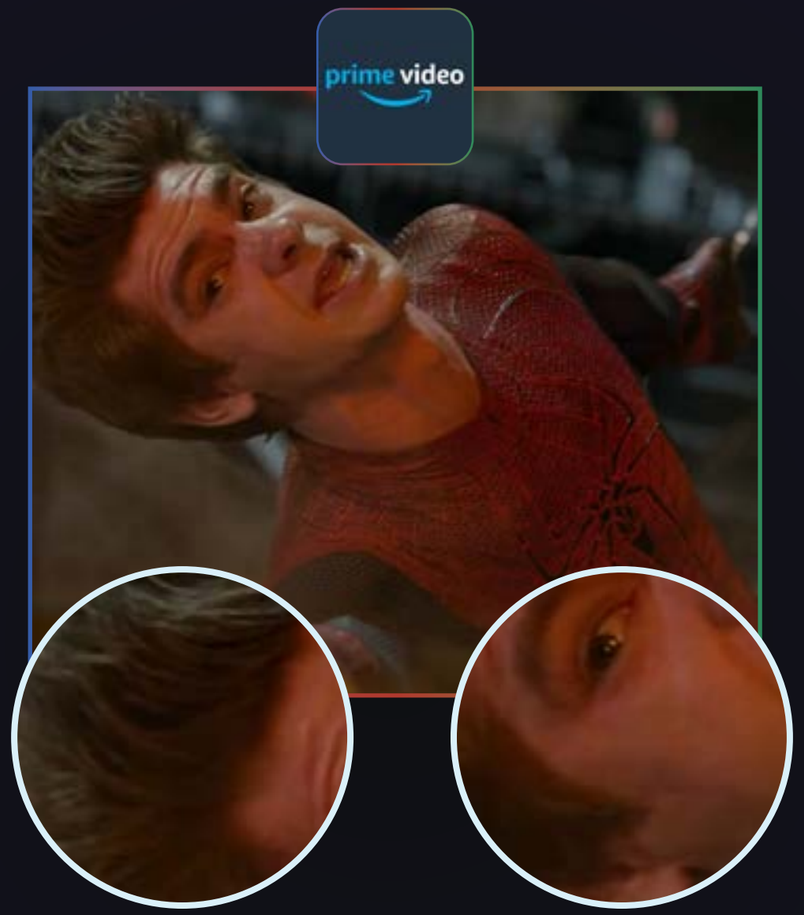
**Playback features**

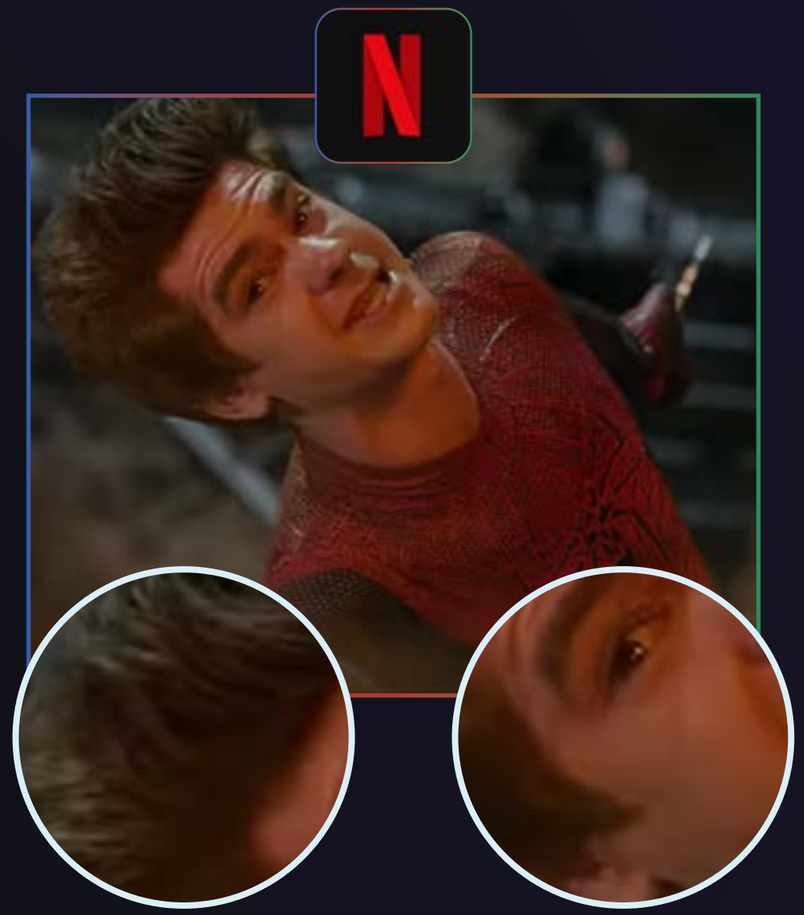
- X-Ray: information on the cast.
- Lightening Bolt: Optimized streaming

**Accessibility**

- AD: Audio description
- CC: Closed Caption

# VIDEO: W/ LEVONO




## Disney+

Disney+ delivered the strongest video quality overall, even when streaming The Amazing Spider-Man under poor internet conditions. The image remained crisp and clear, making it easy to pause the movie and capture clean screenshots. Compared to the other platforms, Disney+ consistently preserved fine detail. In the close-up images of the hair and face, very little blocking or compression artifacting was visible. The image retained sharp edges and smooth gradients, resulting in the highest perceived visual quality among the three platforms.

## Amazon Video

Amazon Prime Video showed the second-best performance. Although its quality was not as good as Disney+, the resolution remained strong and visually clear. In some areas, Prime Video even displayed slightly more pronounced shadows and highlights, giving the image a higher sense of contrast. The video also appeared slightly larger during playback, which may have contributed to the minor reduction in sharpness. Despite this, Amazon Prime Video still maintained good image fidelity and performed well overall.

## Netflix

Netflix showed the most noticeable reduction in visual quality. Blocking and compression artifacts were clearly visible, particularly when pausing the movie to capture screenshots. Many frames appeared blurred at the moment of pause, making it difficult to obtain a clean still image. In the close-up comparisons, facial features and hair lacked clarity, especially along edges, where detail was significantly reduced. This decrease in quality is likely due to Netflix's adaptive streaming compression, which lowers resolution during poor network conditions to prevent buffering and maintain uninterrupted playback.

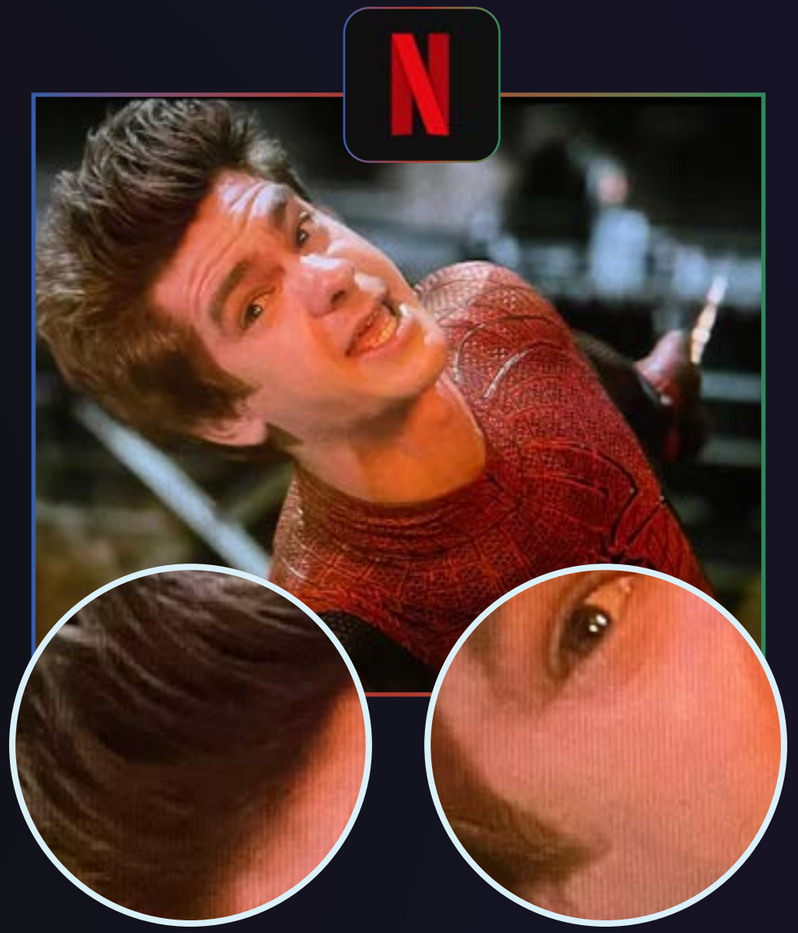
# VIDEO: W/ PIONEER

## Disney+

Similar to the laptop comparison, Disney+ displays strong image quality on the television as well. Fine details in the face and hair remain sharp, and the overall image appears crisp and evenly compressed. This consistency across both devices suggests that Disney+ uses a high-quality compression workflow for The Amazing Spider-Man. Because Disney collaborates closely with Sony, the film's distributor, Disney+ may have access to a higher-quality master or more optimized encoding for this title compared to other platforms.

## Amazon Video

Amazon Prime Video performs well on the television but does not consistently match the sharpness seen on Disney+. While fine facial details are slightly less defined, Prime Video displays stronger contrast, making shadows and highlights more pronounced. This gives the image a slightly different visual character. If Amazon exclusively owns distribution rights to a film, its internally encoded versions may appear at a higher quality than the same film on competing platforms.



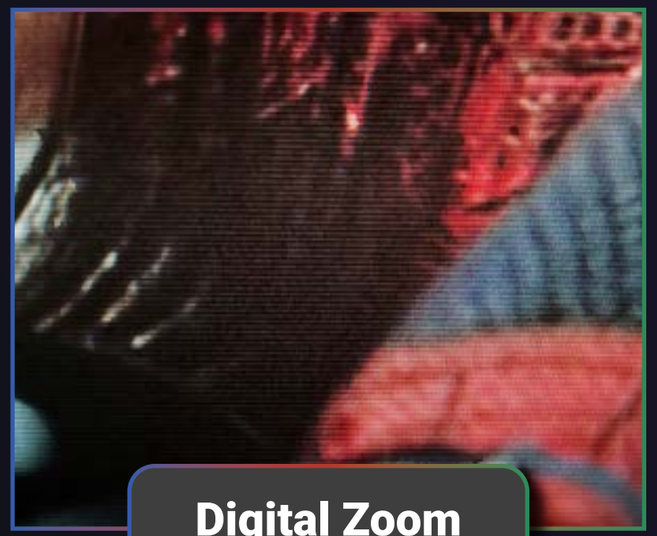
## Netflix

When viewed on the TV, Netflix shows improved quality compared to the laptop display, likely due to the performance of the internet. However, even at this improved level, the image does not reach the crispness of Disney+ or Amazon Video. In close-up views, edges still appear softened and fine detail is reduced, especially around facial features and hair. While these differences are subtle from a distance, they become increasingly apparent in close-up views.

# VIDEO: CLOSER LOOK

Original Frame

Cropped Detail

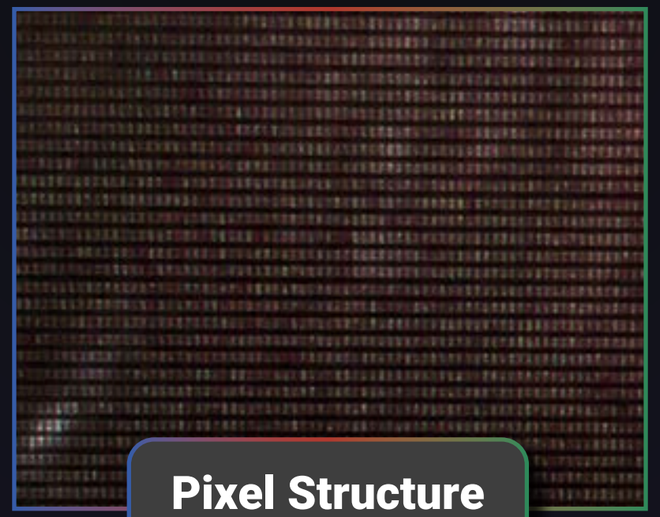


Digital Zoom

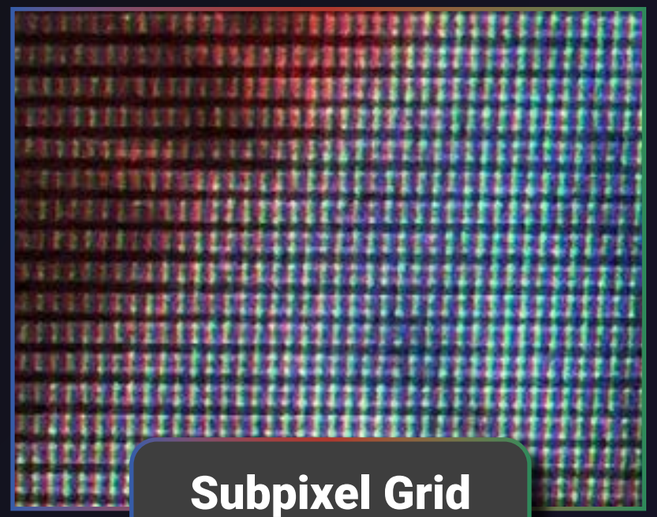
## A Closer Look

When viewing an image on a laptop, individual pixels are difficult to distinguish due to the smaller screen size and higher pixel density. However, when the same image is displayed on a television, the pixels become more visually pronounced as the image is spread across a much larger surface area. This sequence begins with a distant view of the television, followed by progressive zooms that gradually reveal more detail. With each step, the structure of the image becomes clearer until the individual pixels are fully visible.

Once exposed, these pixels appear as small square units, with each square representing a single color value. Together, millions of these colored squares blend to form the complete image. This comparison demonstrates how resolution, screen size, and viewing distance directly affect how clearly pixel structure can be perceived.



Pixel Structure



Subpixel Grid

# Asset Management

# Summary

Throughout this project, all digital assets were organized using a structured naming and folder system to preserve original files while tracking multiple compressed outputs. Each image, audio, and video asset included identifying information such as subject name, compression tool used (Photoshop, Squoosh, GIMP, or platform capture), quality level, and file format directly within the filename. This ensured that each test version could be accurately compared without confusion or data loss.

Original files were kept separate from exported and compressed versions to prevent overwriting and to maintain source integrity. Multiple quality levels were exported for each format to allow side-by-side visual and file-size comparison. This versioned asset management workflow allowed for efficient testing, repeatable compression experiments, and clear documentation of how quality, format, and platform streaming affected final media output.

# Conclusion

Through this investigation, I gained a much deeper understanding of how compression directly affects image, audio, and video quality across different platforms and applications. By testing multiple formats such as JPEG, WebP, AVIF, MP3, WAV, FLAC, HEVC, and AV1, I learned how lossy compression reduces file size by removing visual or auditory data, while lossless compression preserves original quality at the cost of larger file sizes. I also observed how bit depth, resolution, and encoding methods influence clarity, artifacting, and overall media performance. Seeing these changes occur firsthand made the technical concepts behind compression much more tangible and meaningful.

Understanding compression processes, technologies, and the issues associated with them is essential in today's digital media industry. Designers, developers, and content creators must constantly balance visual quality, file size, and performance—especially for web-based environments where loading speed and compatibility play a critical role in user experience. Improper compression choices can lead to degraded media, inefficient storage, and poor usability, while informed decisions can greatly enhance both aesthetic quality and functional performance.

This project has strengthened my ability to evaluate media formats critically and apply compression tools with intention rather than guesswork. The knowledge gained from this investigation will directly influence my future work in digital design, web development, and media production. Moving forward, I will be more confident in selecting appropriate formats, optimizing assets for performance, and maintaining professional standards for media quality in both academic and professional settings.

# GLOSSARY

Aliasing – A visual distortion that occurs when an image or line is displayed at too low a resolution, creating jagged, stair-step edges.

Anti-Aliasing – A smoothing technique that blends edge pixels with surrounding colors to reduce the appearance of jagged edges.

Aspect Ratio – The proportional relationship between an image's width and height (such as 16:9 or 4:3).

AVIF – A modern image format that provides high compression efficiency and strong image quality at small file sizes.

Bit Depth – The number of bits used to represent the color of a single pixel, affecting how many colors can be displayed.

Codec – A method used to compress and decompress digital media files.

Compression – The process of reducing file size by removing or simplifying data.

Lossless Compression – Compression that preserves all original data without reducing quality.

Lossy Compression – Compression that removes some data to reduce file size, resulting in a loss of quality.

JPEG – A commonly used lossy image format known for small file sizes but prone to compression artifacts.

PNG – A lossless image format that supports transparency and high image quality.

WEBP – A modern image format developed by Google that supports both lossy and lossless compression.

Pixel – The smallest unit of a digital image, representing a single color value.

PPI (Pixels Per Inch) – A measurement of pixel density that affects image sharpness on a display.

Raster Image – An image made from a grid of pixels; quality decreases when scaled up.

Vector Image – An image created using mathematical paths that can scale infinitely without losing quality.

Resolution – The total number of pixels in an image or display, typically written as width × height.

Sample Rate – The number of audio samples captured per second in a digital recording.

Bitrate – The amount of data processed per second in audio or video files.

Artifacting – Visual distortions caused by heavy compression.

Streaming – The real-time delivery of digital media over the internet without full file downloading.

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